PIXORA – SOCIAL MEDIA PLANNING AND APPROVAL PLATFORM

A PROJECT REPORT

Submitted by

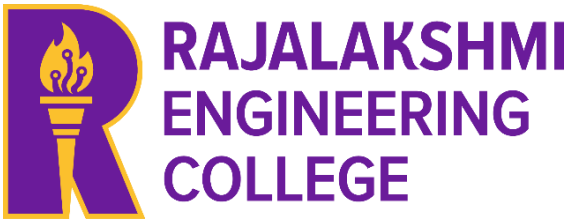
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in partial fulfilment of the End Semester Examination for the course

CB23F35 – LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING
COMPUTER SCIENCE AND BUSINESS SYSTEMS



RAJALAKSHMI ENGINEERING COLLEGE
(An Autonomous Institution, Affiliated to Anna University, Chennai)
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NOVEMBER 2026

RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled **"PIXORA – SOCIAL MEDIA PLANNING AND APPROVAL PLATFORM"** is the bonafide work of **SHARMILA G (241401097),** who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027,Semester V, under my supervision.

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DECLARATION

I, SHARMILA G (241401097), hereby declare that the project report titled "PIXORA – SOCIAL MEDIA PLANNING AND APPROVAL PLATFORM "submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date:

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SHARMILA G

ABSTRACT

Social media plays an important role in promoting college events, activities and achievements. College clubs and departments regularly create posters, captions and promotional content to share on platforms such as Instagram. However, managing these activities through messaging groups and spreadsheets can become difficult. Assigning tasks, collecting content, reviewing designs, obtaining approvals and maintaining publishing schedules are often handled manually. This may result in communication gaps, delayed approvals, missed deadlines and difficulty in tracking content performance.

This project presents PIXORA – Social Media Planning and Approval Platform, developed using the Zoho Creator low-code application development platform. The application provides a centralised workspace for managing the social media content workflow of college clubs and departments. It includes nine forms covering club registrations, team members, content ideas, content tasks, content submissions, content reviews, publishing schedules, published posts and engagement tracking. A role hierarchy of Admin, Team Lead and Team Member is used to manage responsibilities. Zoho Creator forms, reports, a dashboard, Deluge scripting and workflows are used to organise information, update statuses automatically and support the approval process. An integration with Zoho WorkDrive for storing creative files has been set up and is partially implemented.

The application is designed to simplify collaboration between team members and to improve the overall content management process. Basic record creation and some status automation have been tested, and sample data for a college Film Club is being used to verify the dashboard, charts and reports. Role-based access, report filters and automatic WorkDrive file upload are still to be fully verified. PIXORA records publishing details but does not publish posts automatically; future enhancements include direct Instagram and Facebook integration, AI-powered caption generation and automatic engagement tracking.

Keywords: Low-Code Development; Zoho Creator; Social Media Management; Workflow Automation; Content Approval; Deluge Scripting

PROJECT AT A GLANCE

Attribute	Details
Project Title	PIXORA-Social Media Planning and Approval Platform
Application Domain	Social Media Management
Student Name & Register No.	SHARMILA G - 241401097
Zoho Creator Edition	Trial edition
Application Name (Link Name)	PIXORA
Published / Live URL	https://creatorapp.zoho.in/sharmila.g.2024.csbs_raja613/social-media-content-management-system
Demo Credentials (per role)	Admin, Team Lead, Team Member (Demo login credentials to be provided separately)
Number of Forms / Subforms	9 Forms / 0 Subforms
Number of Reports (by type)	Approximately 12 Reports: List Reports – 12 (type-wise count to be verified from the application)
Number of Pages / Dashboards	6 Planned-1 main Dashboard and 5 Custom Pages
Number of Workflows	Form: 3 configured; Schedule: Planned; Approval: Planned; Blueprint: 0; Batch: 0
Deluge Functions / Custom Actions	2 verified Deluge workflow scripts (DS-01 and DS-03) and custom function uploadtoworkdrive (WorkDrive integration created, end-to-end operation not verified)
Integrations	Zoho WorkDrive (zoho_workdrive_connection) – Connection status: Connected; integration partially implemented
Roles & Profiles Defined	Admin, Team Lead, Team Member (role hierarchy)
Environments Used	Development, Stage and Production (listed in Zoho Creator; deployment across environments not verified)
Date of Production Deployment	26 Oct,2026

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CSV	Comma-Separated Values
ER	Entity–Relationship
FR	Functional Requirement
KPI	Key Performance Indicator
MVP	Minimum Viable Product
NFR	Non-Functional Requirement
RBAC	Role-Based Access Control
SaaS	Software as a Service
UAT	User Acceptance Testing
UI	User Interface
URL	Uniform Resource Locator
Zia	Zoho's AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

Social media plays an important role in promoting college events, activities and achievements. College clubs regularly use platforms like Instagram to share posters, captions and event updates.

Managing these activities manually through WhatsApp groups and spreadsheets can be difficult. Assigning tasks, collecting posters, getting approvals and maintaining posting schedules may lead to confusion and delays.

To solve these problems, PIXORA – Social Media Planning and Approval Platform is developed using Zoho Creator. It provides a centralised platform to manage content ideas, assign tasks, submit posters and captions, approve content, schedule posts and track engagement.

The application includes three main roles: Admin, Team Lead and Team Member. It helps improve teamwork and makes social media content management easier and more organised.

1.2 Problem Statement

College clubs manage their social media activities through different messaging groups and manual methods. This makes it difficult to manage tasks, content submissions, approvals and publishing schedules in one place.

The major problems are:

- Manual task assignment makes tracking team activities difficult.
- Content approval may be delayed due to poor communication.
- Content ideas and submissions are stored in different places.
- Publishing schedules are difficult to maintain.
- Post engagement details are not tracked systematically.

1.3 Objectives

The main objectives of PIXORA are:

• To develop a centralised social media management application using Zoho Creator.

• To manage clubs, team members and content ideas.

• To assign tasks and track their progress.

• To manage poster and caption submissions with an approval process.

• To maintain publishing schedules and published post records.

• To track social media engagement through reports and dashboards.

1.4 Scope of the Project

In-Scope:

• Club and team member management.

• Content idea creation and task assignment.

• Poster and caption submission.

• Content review and approval.

• Social media publishing schedules.

• Published post management and engagement tracking.

Out-of-Scope:

• Automatic publishing to Instagram and other platforms.

• Online advertisement payments.

• Advanced audience prediction and sentiment analysis.

• Direct integration with social media APIs.

1.5 Justification for a Low-Code Approach

Zoho Creator was selected as the development platform for PIXORA – Social Media Planning and Approval Platform because it provides a low-code environment that simplifies application development and reduces the need for extensive programming. Its built-in features and customisation capabilities make it suitable for developing an application that manages social media content planning, task assignments, creative submissions, approval processes and publishing schedules.

The major advantages of using Zoho Creator are explained below:

1. **Easy Application Development:** Zoho Creator provides drag-and-drop tools and ready-made components that simplify the creation of forms, pages and application interfaces with minimal coding.
2. **Efficient Data Management:** Its form and report features help store, organise, retrieve and manage information related to clubs, team members, content ideas, tasks, submissions and published posts.
3. **Workflow Automation:** Zoho Creator supports Deluge scripting, which enables the automation of activities such as task status updates, member notifications and content submission processes.
4. **User Role and Access Management:** The platform provides role-based access control, allowing administrators, team leads and team members to access application features according to their assigned permissions.
5. **Dashboard and Reporting:** Built-in reports, KPI cards and charts provide a clear view of content activities, task progress, approval status and publishing information.
6. **Integration Capabilities:** Zoho Creator supports integrations with other services, such as Zoho WorkDrive, helping manage and access creative files within the application.
7. **Reduced Development Effort:** The availability of built-in components, automation tools and integration features reduces the amount of manual coding and simplifies application maintenance.

Therefore, Zoho Creator is an appropriate platform for developing PIXORA, as it provides the necessary tools to build, manage and automate social media planning and approval activities efficiently with reduced development complexity.

1.6 Organisation of the Report

This report is organised into nine chapters, each describing a different phase of the development and implementation of the PIXORA – Social Media Planning and Approval Platform.

Chapter 1 – Introduction: Introduces the project, its background, objectives, scope, problem statement and significance.

Chapter 2 – Existing and Proposed System: Explains the existing social media management process, its limitations and the proposed PIXORA system with its key features and advantages.

Chapter 3 – System Requirements: Describes the functional and non-functional requirements, hardware and software requirements, and the roles and responsibilities of different users.

Chapter 4 – System Analysis and Design: Presents the system architecture, workflow diagrams, use case diagrams, database design, entity relationship diagram and user interface design.

Chapter 5 – System Implementation: Explains the implementation of PIXORA using Zoho Creator, covering forms, reports, dashboards, workflows, automation, integrations and access permissions.

Chapter 6 – Testing and Debugging: Discusses the testing approach, test cases, test results, debugging activities and user acceptance testing performed to evaluate the functionality and reliability of the application.

Chapter 7 – Deployment and Performance Optimisation: Describes the application deployment process, deployment checklist, security measures, access control and performance optimisation techniques.

Chapter 8 – Results and Discussion: Presents the application outputs, dashboard views, implemented features and overall results achieved through the PIXORA platform.

Chapter 9 – Conclusion and Future Enhancements: Summarises the project outcomes, highlights the benefits of PIXORA and discusses possible future improvements and additional features.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

At present, the social media activities of college clubs are handled through several unconnected tools such as WhatsApp, Google Drive, spreadsheets and Instagram. Typically, content ideas are discussed in WhatsApp groups, tasks are assigned through messages, creatives are shared through chat or Google Drive, approval is requested in chat, publishing dates are noted in spreadsheets and the post is published manually on Instagram. Figure 2.1 shows this as-is process.

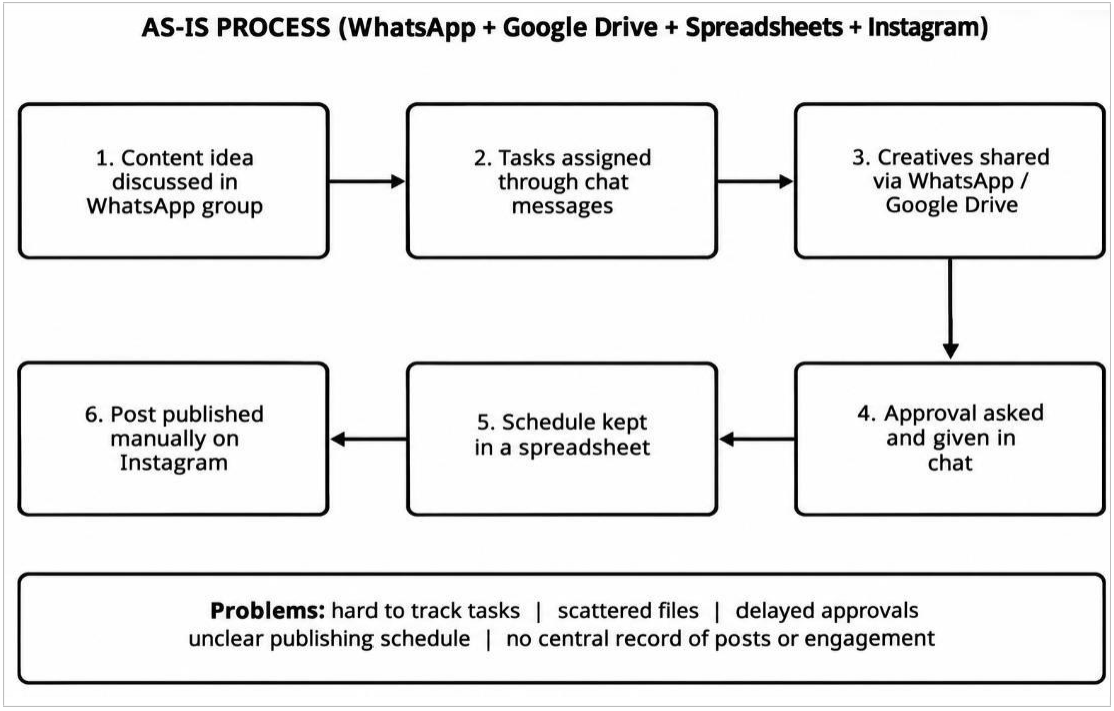


Figure 2.1: As-Is Process Flow of the Existing System

2.2 Limitations of the Existing System

- Task assignment is difficult and coordination between team members is poor.
- Creative submissions are difficult to track because files are spread across chat groups and drives.
- Content approval is delayed because there is no structured review process.
- Publishing schedules are unclear, which causes confusion about what is to be posted and when.

- There is no proper record of published posts and no central way to monitor social media activities or engagement.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Longer: database, backend, user interface and hosting have to be built separately.	Shorter: forms, reports, dashboard and workflows are configured on the platform.
Skills required	Programming, database and web-development skills.	Data modelling and basic Deluge scripting; suitable for a student developer.
Hosting & infrastructure	Servers and hosting must be arranged and maintained.	Hosted and managed by Zoho Creator.
Mobile application	Separate mobile app development.	Zoho Creator mobile app is available (not tested for PIXORA).
Security & access control	Authentication and role logic must be coded.	Built-in users, roles and permission sets (Admin, Team Lead, Team Member).
Maintenance & change	Code changes, testing and redeployment for each change.	Forms, fields, reports and workflows can be changed in the builder.
Customisation limits	Full control over design and logic.	Limited to platform features; for example, direct Instagram publishing needs API-level (pro-code) work.
Cost	Developer effort, hosting and licences.	Platform subscription; no separate infrastructure.

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	W. Rosam and O. Kalnichevski [1]	Low-Code Development: A Practical Guide (BPB Publications)	Explains low-code concepts and how applications can be built with little hand-written code.	Course reference for the low-code approach used in PIXORA.
2	Zoho Corporation [2]	Zoho Creator Help Documentation	Describes forms, reports, pages, workflows, connections and user roles in Zoho Creator.	Basis for the nine forms, reports, dashboard and roles.
3	Zoho Corporation [3]	Deluge Script Reference Guide	Documents Deluge syntax, variables and scripting in forms and workflows.	Used to understand the Deluge script that sets submission status and date.
4	Zoho Corporation [4]	Zoho WorkDrive Help Documentation	Explains file storage, folders and sharing in Zoho WorkDrive.	Reference for storing creative files in the PIXORA_creatives folder.
5	Gartner, Inc. [5]	Magic Quadrant for Enterprise Low-Code Application Platforms	Market analysis of low-code application platform vendors and their capabilities.	Shows the industry relevance of low-code platforms such as Zoho Creator.
6	A. C. Bock and U. Frank [6], 2021	Low-Code Platform (Business & Information Systems Engineering)	Discusses what low-code platforms are and how they differ from traditional development.	Supports the justification for a low-code approach (Section 1.5).
7	A. Sahay et al. [7], 2020	Supporting the Understanding and Comparison of Low-Code Development Platforms	Presents a feature-based comparison of low-code development platforms.	Helps in understanding platform features such as forms, workflows and integration.

2.5 Emerging Trends in Low-Code Development

Current trends in low-code development include citizen development, where users with limited programming experience build business applications; AI-assisted application

building; hyper-automation, where workflows connect many business steps; and composable applications that combine services through integrations [5], [7].

These trends influenced PIXORA in the following ways. The application was built by a student developer using drag-and-drop forms, reports and pages, which reflects citizen development. Workflows and Deluge scripts automate repetitive steps such as setting statuses. The Zoho WorkDrive integration follows the composable approach of connecting services. AI-powered caption generation and recommendations are not part of the current version and are kept as future enhancements.

2.6 Proposed System

PIXORA brings the complete content workflow into one Zoho Creator application: Content Idea → Task Assignment → Creative Preparation → Content Submission → Content Review → Approval or Rejection → Publishing Schedule → Published Post → Engagement Tracking. It overcomes the limitations in Section 2.2 as follows:

- Task assignment and coordination: the Content Tasks form lets Team Leads assign tasks with priority, due date and status, and a workflow notifies the assigned member.
- Submission tracking: the Content Submissions form collects creative files with submission status and date.
- Approval: the Content Reviews form provides a structured approve or reject decision with feedback.
- Publishing schedule: the Schedules form maintains Draft, Scheduled and Published statuses.
- Records and monitoring: the Published Posts and Engagement Trackers forms, together with dashboard KPI cards and charts, give a central view of activity.

The current MVP records publishing information; it does not publish posts automatically to Instagram or other platforms.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Admin	Manages the overall application and its configuration.	Manage team members and configurations, access application records, monitor overall activities.
Team Lead	Leads the social media team and controls the approval process.	Assign tasks, monitor team progress, review submissions, approve or reject content, manage publishing activities.
Team Member	Designers, content creators and social media managers working on assigned content.	View assigned tasks, create or update content as permitted, submit creative files, track own work.

Role hierarchy in Zoho Creator: Admin → Team Lead → Team Member. The team_role field in the Team Members form stores job roles such as Admin, Team Lead, Designer, Content Creator and Social Media Manager.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall maintain club and team member records, including team role and contact number.	Admin	High
FR-02	The system shall allow users to create and maintain content ideas with platform, content type, status and related club.	Team Lead	High
FR-03	The system shall allow Team Leads to create content tasks and assign them to team members with priority and due date.	Team Lead	High
FR-04	The system shall set a new task's status to Pending automatically and notify the assigned member.	System	Medium
FR-05	The system shall allow assigned team members to submit creative files for a task.	Team Member	High
FR-06	The system shall set the submission status to Submitted and record the submission date automatically.	System	High
FR-07	The system shall allow Team Leads to review submissions and record approval or rejection with feedback.	Team Lead	High
FR-08	The system shall maintain the publishing schedule with Draft, Scheduled and Published statuses.	Team Lead	Medium

FR ID	Requirement	Role(s)	Priority
FR-09	The system shall record published posts with platform, published date and post URL.	Team Lead	Medium
FR-10	The system shall record likes, comments, shares and reach for published posts.	Team Lead	Medium
FR-11	The system shall display a dashboard with KPI cards, charts and reports.	Admin, Team Lead	Medium
FR-12	The system shall restrict access according to the roles Admin, Team Lead and Team Member.	Admin	High
FR-13	The system shall store creative files in Zoho WorkDrive (partially implemented).	Team Member	Low

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports and the dashboard shall load without noticeable delay for the sample data volume used in the project.
NFR-02	Security	Users shall access only the data permitted by their role (final verification pending).
NFR-03	Usability	The application shall be usable through a standard web browser, with quick-action shortcuts on the dashboard.
NFR-04	Availability	The application shall be available online through the Zoho Creator cloud platform.
NFR-05	Maintainability	Forms, workflows and reports shall be modifiable in Zoho Creator without rebuilding the application; consistent field link names shall be used.
NFR-06	Auditability	Record changes shall be traceable using Zoho Creator's audit and record history features (to be verified).

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
API testing tool	Not used; the WorkDrive connection was checked using the connection test in Zoho Creator
Other Zoho services	Zoho WorkDrive (folder: PIXORA_creatives)
Mobile	Not tested in this project
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

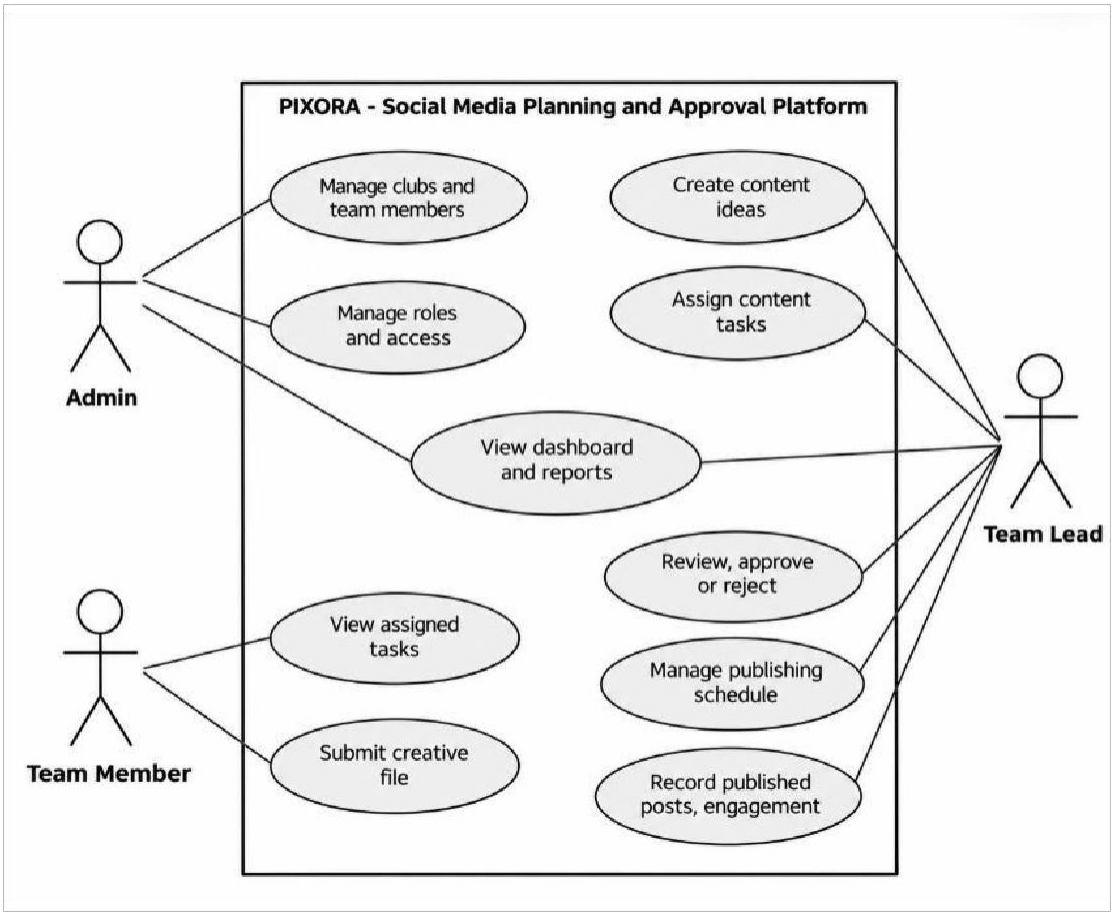


Figure 3.1: Use Case Diagram of the Application

Table 3.5: Use Case Description – Review and Approve Content Submission

Use Case ID	UC-01
Use Case Name	Review and Approve Content Submission
Actor(s)	Team Lead
Pre-condition	A team member has submitted a creative through Content Submissions (submission status = Submitted).
Main Flow	1. Team Lead opens Content Reviews. 2. Team Lead selects the submitted creative and checks it. 3. Team Lead sets the review status to Approved. 4. Team Lead saves the review record.
Alternate Flow	Team Lead sets the review status to Rejected and enters feedback; the team member revises and resubmits the creative.
Post-condition	The review record is stored with status, feedback and review date; approved content can be scheduled for publishing.
Related FR	FR-05, FR-07, FR-08

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

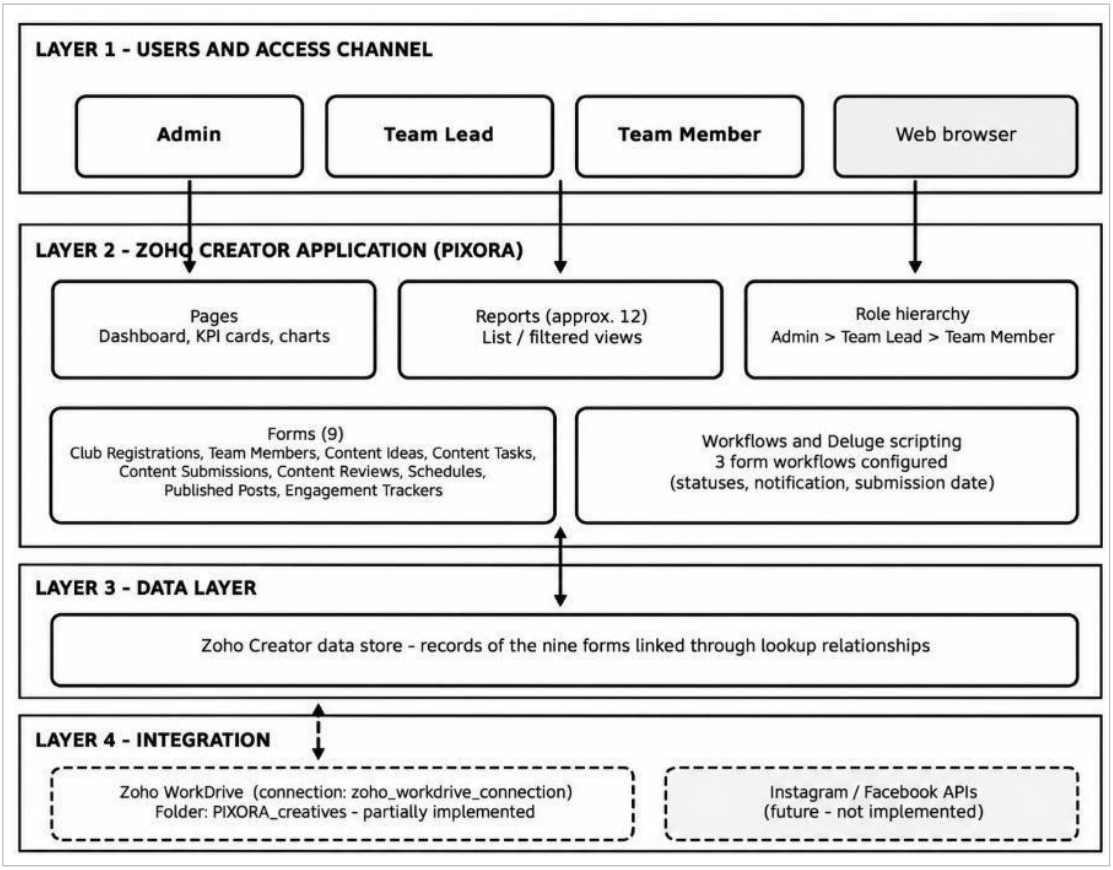


Figure 4.1: System Architecture of the Application

PIXORA follows a four-layer architecture, shown in Figure 4.1.

- Users and access channel: Admin, Team Lead and Team Member use PIXORA through a web browser.
- Zoho Creator application: pages (dashboard), reports, nine forms, workflows and Deluge scripts, controlled by the role hierarchy.
- Data layer: records of the nine forms are stored in Zoho Creator and linked through lookup relationships.
- Integration: Zoho WorkDrive is connected through zoho_workdrive_connection (partially implemented). Direct Instagram and Facebook API integration is a future enhancement.

4.2 Data Model

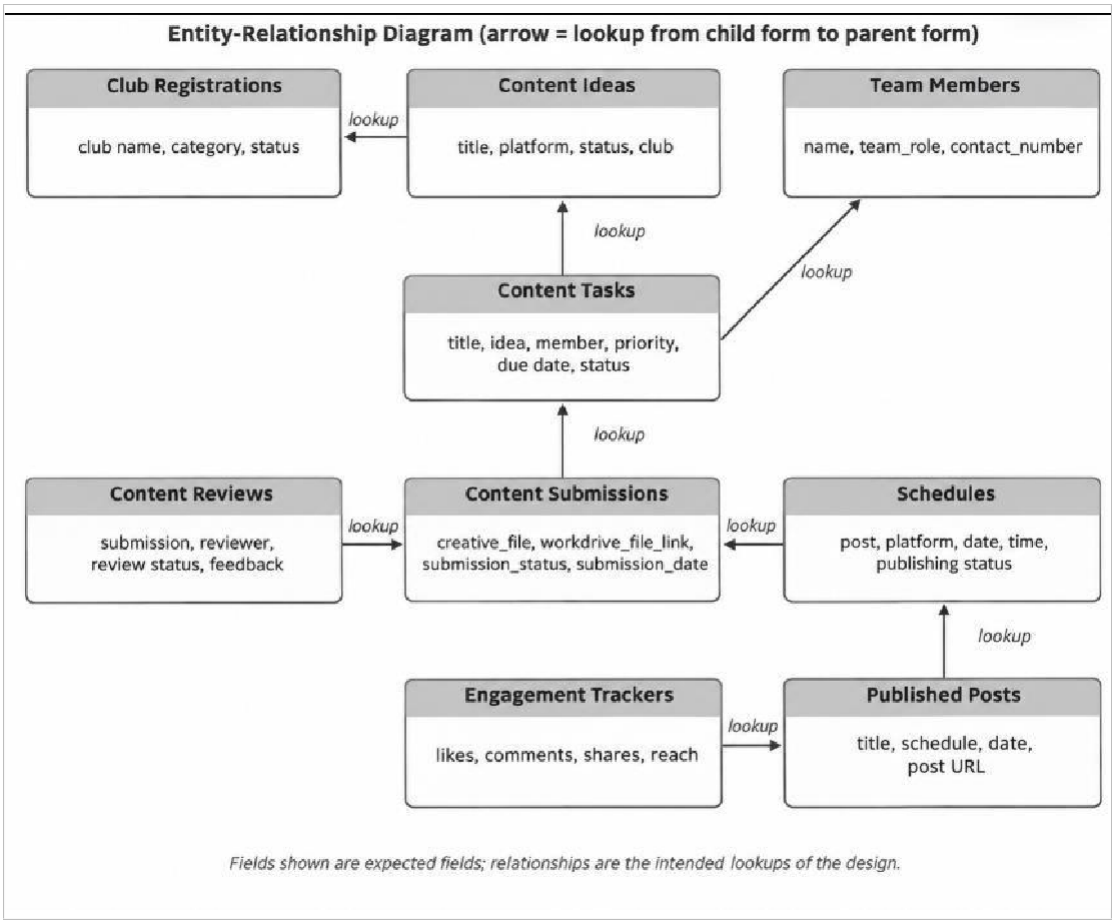


Figure 4.2: Entity–Relationship Diagram

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Club Registrations	Club name, category, description, registration details, status	Content Ideas	One-to-Many	Referenced by Content Ideas
Team Members	Member name, email, department, team_role, contact_number, status	Content Tasks	One – to-Many	Referenced by Content Tasks
Content Ideas	Idea title, platform, content type, description, status, club	Club Registrations	Many-to-One	Lookup field for the club
Content Tasks	Task title, idea, assigned member, priority, due date, status	Content Ideas; Team Members	Many-to-One	Lookup fields for idea and assigned member

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Content Submissions	creative_file, workdrive_file_link, submission_status, submission_date	Content Tasks	Many-to-One	Lookup field for the task
Content Reviews	Submission, reviewer, review status, feedback, review date	Content Submissions	Many-to-One	Lookup field for the submission
Schedules	Post title, platform, scheduled date and time, publishing status	Content Submissions	Many-to-One	Lookup field for the submission
Published Posts	Post title, platform, schedule, published date, post URL, status	Schedules	Many-to-One	Lookup field for the schedule
Engagement Trackers	Published post, likes, comments, shares, reach, date	Published Posts	Many-to-One	Lookup field for the post

Note: contact_number, team_role, creative_file, workdrive_file_link, submission_status and submission_date are confirmed field link names. The other fields and the lookup configuration are the intended design and are to be verified againstthe live application.

4.3 FORM DESIGN

Table 4.2: Field Specification – Team Members and Content Submissions Forms

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Creative File	creative_file	File Upload	yes	No specific validation configured	Stores the creative uploaded by the team member
WorkDrive File Link	workdrive_file_link	URL	yes	No specific validation configured	Intended to hold the WorkDrive file URL (automatic upload not yet verified)
Submission Status	submission_status	Dropdown/choice	No	Set to "Submitted" by workflow	Updated by script DS-01

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Submission Date	submission_date	Date	no	Set to today by workflow	Updated by script DS-01
Contact Number	contact_number	Phone	Yes	No additional validation configured	Team Members form
Team Role	team_role	Dropdown/Choice	Yes	Example values: Admin, Team Lead, Designer, Content Creator, Social Media Manager	Team Members form

4.4 Process and Workflow Design

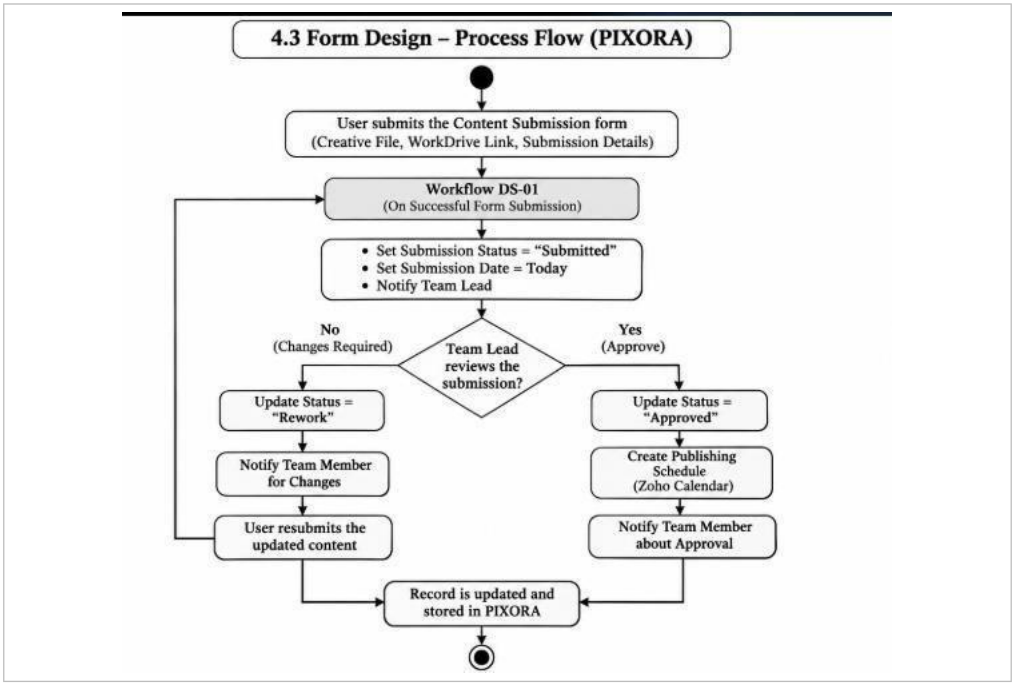


Figure 4.3: Workflow / Process Flow for the Content Approval and Publishing Process

4.5 User Interface and Navigation Design

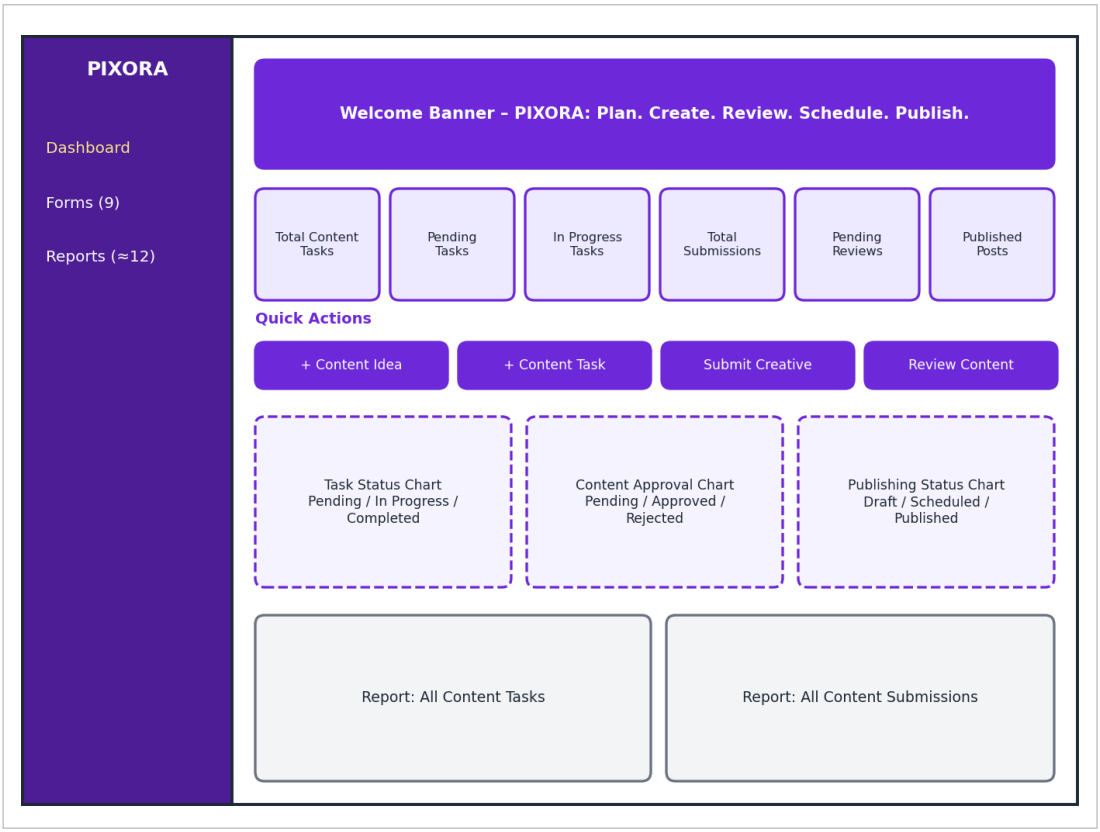


Figure 4.4: Navigation Structure and Wireframe of the Main Dashboard

The wireframe shows the planned layout of the main dashboard: welcome banner, six KPI cards, quick actions, three status charts and two embedded reports, with the navigation menu on the left.

4.6 Access Control Design

Table 4.3: Role–Permission Matrix

Component	Admin	Team Lead	Team Member
Forms: Club Registrations, Team Members	A E V D	V	—
Form: Content Ideas	A E V D	A E V	V
Form: Content Tasks	A E V D	A E V	V (assigned)
Form: Content Submissions	V	V	A E V (own)
Form: Content Reviews	V	A E V	—
Forms: Schedules, Published Posts, Engagement Trackers	V	A E V	—
Page: Dashboard and Reports	V	V	V (limited)

A = Add, E = Edit, V = View, D = Delete, — = No access. The matrix shows the access intended for each role. The final role-to-permission assignments (permission sets Designer, Read, Write and Team Lead Access) are yet to be fully verified.

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

PIXORA was developed in Zoho Creator as an individual low-code project. The application contains nine forms, approximately 12 reports, a dashboard with a professional purple/violet SaaS-style interface, workflows with Deluge scripting, a Zoho WorkDrive connection and a role hierarchy of Admin, Team Lead and Team Member. CSV import was used to prepare sample data, and sample records are also being added manually for dashboard testing.

The application was created using Zoho Creator's AI-assisted builder (Zia), followed by manual configuration and customization of forms, reports, workflows and dashboard components.

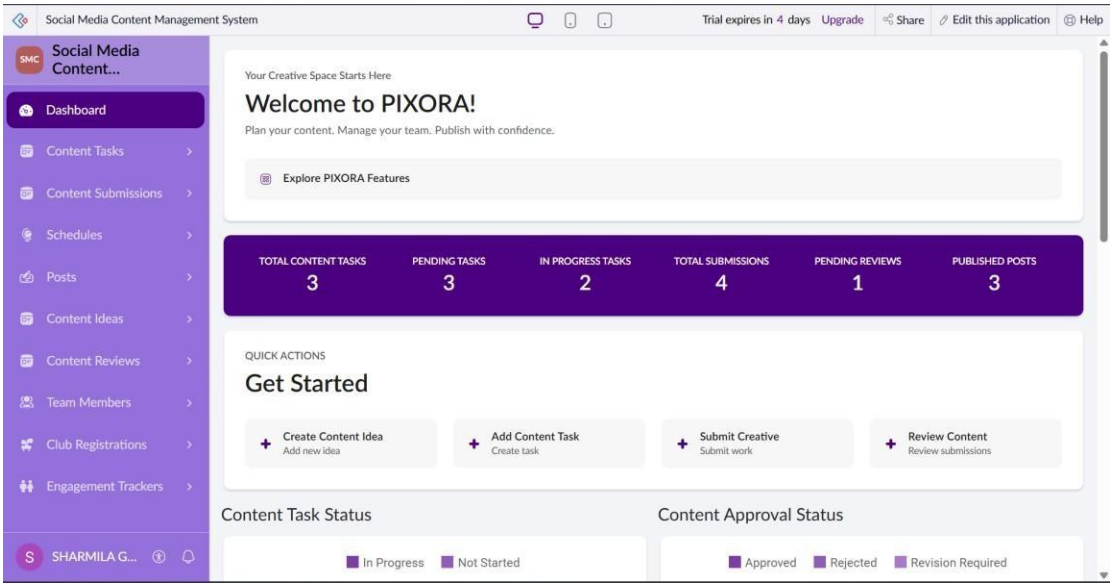


Figure 5.1: Application Home and Navigation Sections

5.2 Forms and Data Capture

PIXORA has nine forms, described in Table 4.1. Basic record creation has been tested for Content Ideas, Content Tasks and Content Submissions.

The Content Submissions form is described in detail because it combines file upload, status automation and the WorkDrive integration. It contains the creative_file (File Upload) and workdrive_file_link (URL) fields, the submission_status and submission_date fields, and task-related information. The Team Members form has the

required fields contact_number and team_role. Field types of the remaining fields, lookup configurations and validations are to be confirmed from the form builder.

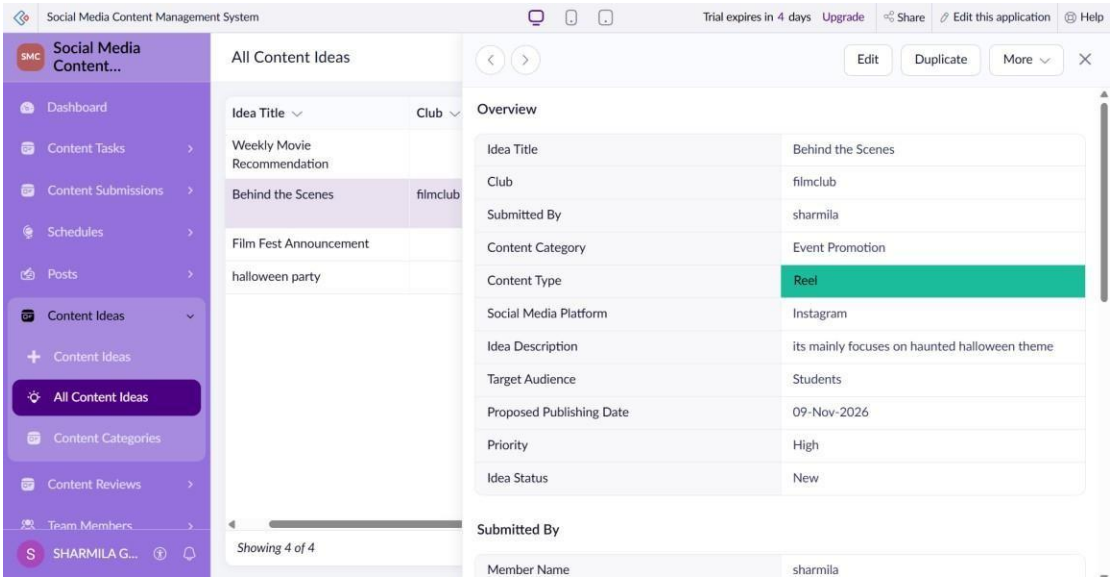


Figure 5.2(a): Content Ideas Form – Live Mode Output

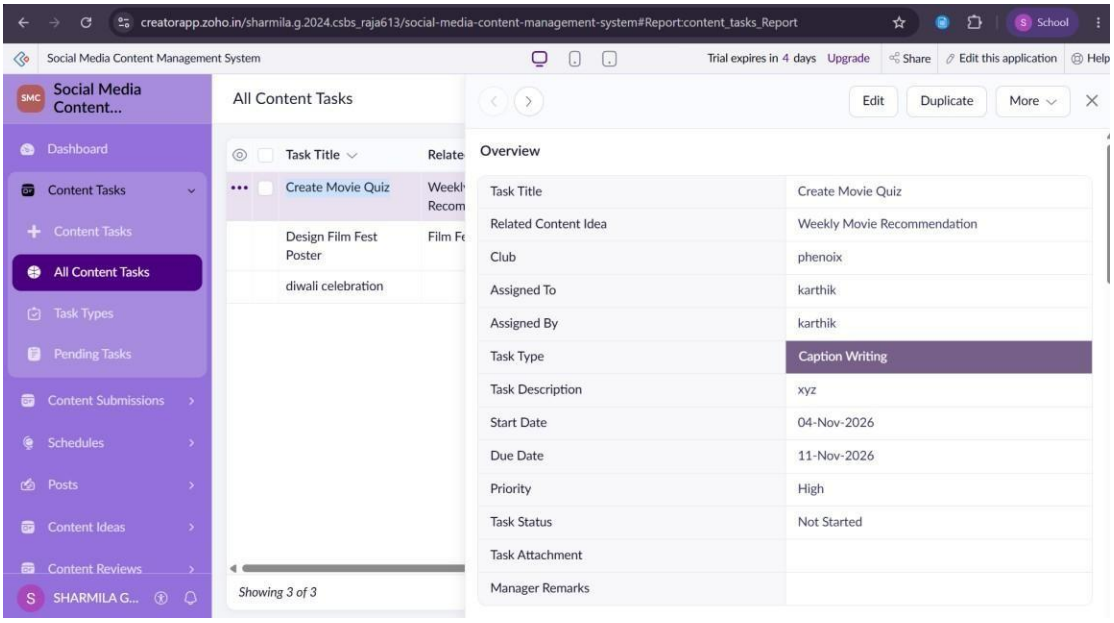


figure5.2(b): Content Tasks Form – Live Mode Output

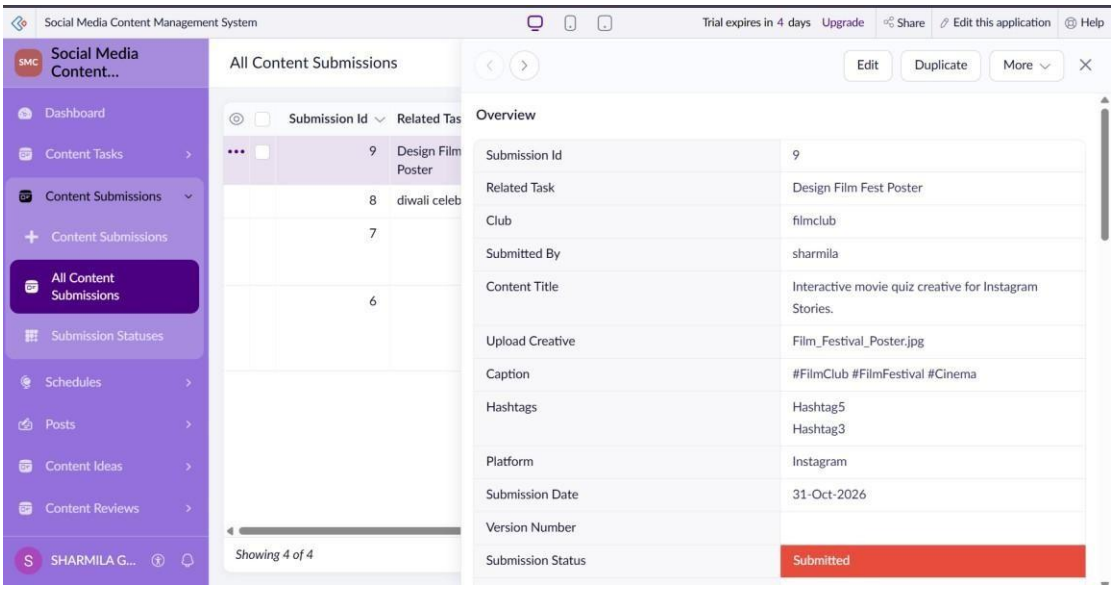


Figure 5.2(c): Content Submissions Form – Live Mode Output

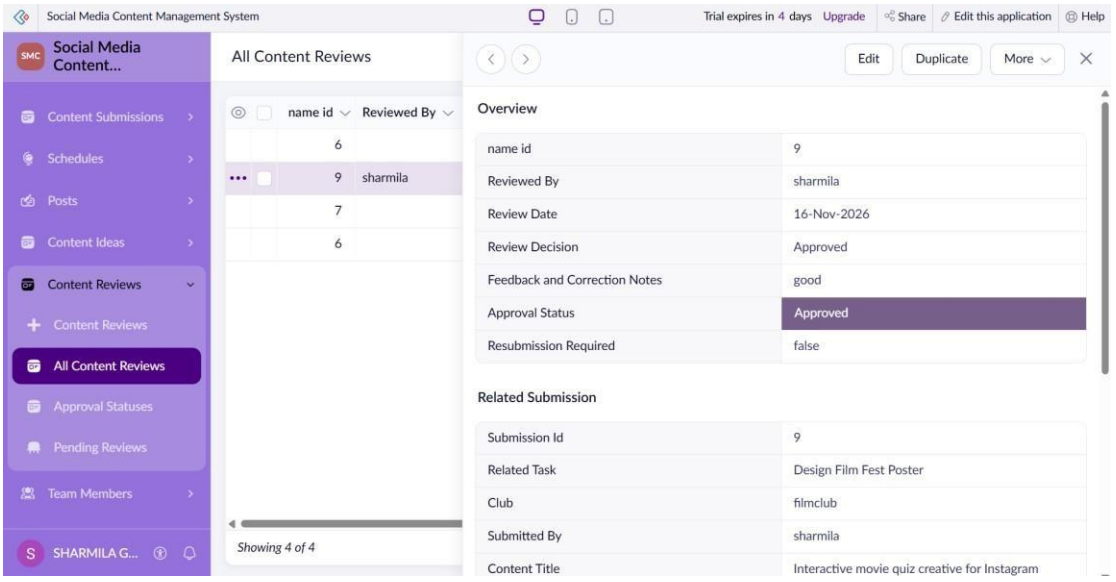


Figure 5.2(d): Content Reviews Form – Live Mode Output

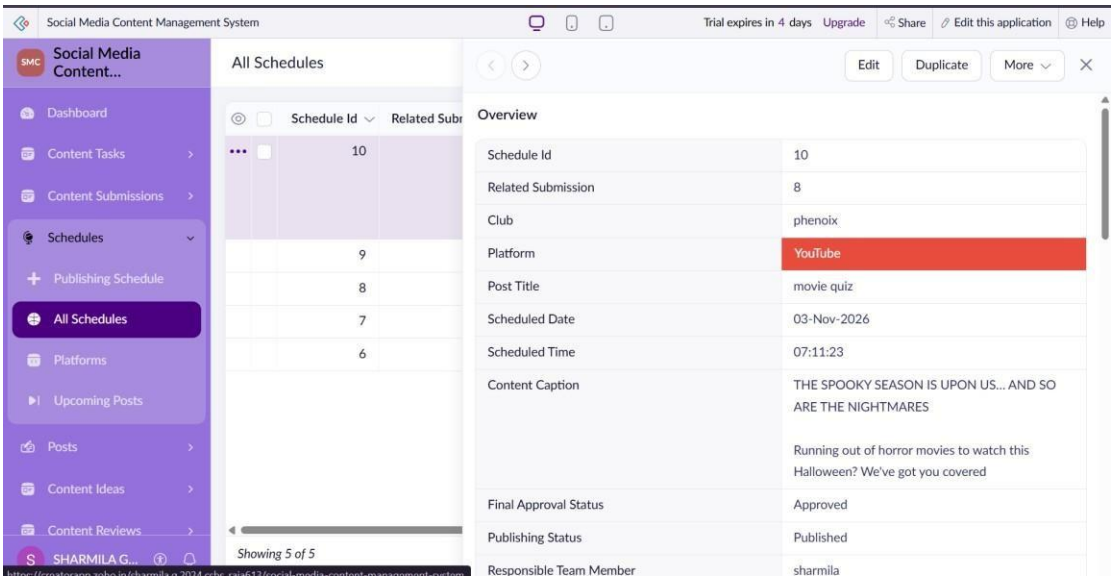


Figure 5.2(e) :- Publishing Schedule Form – Live Mode Output

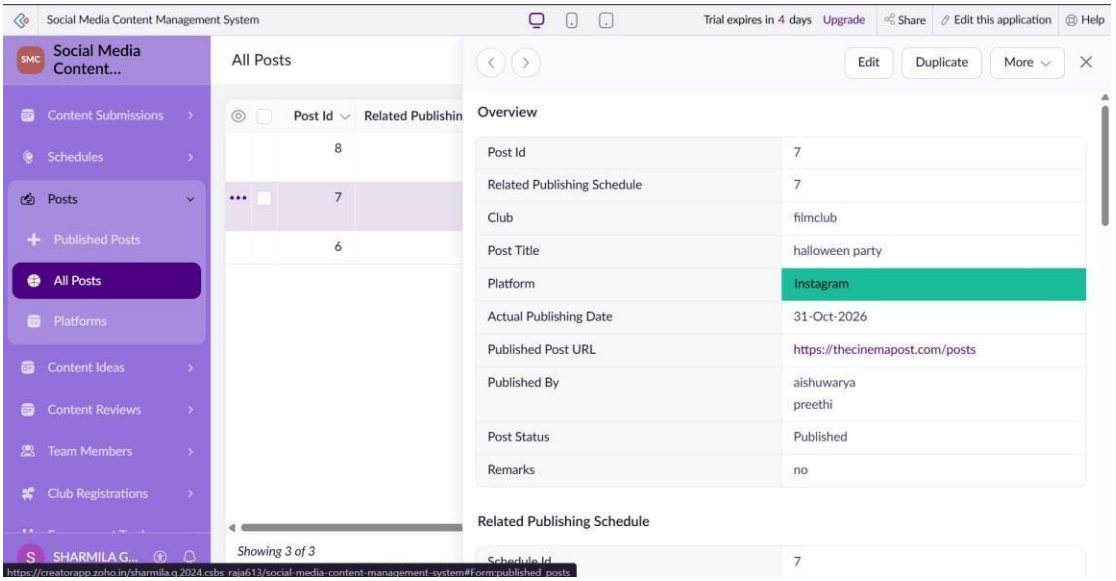


Figure 5.2(f) :- Posts Form – Live Mode Output

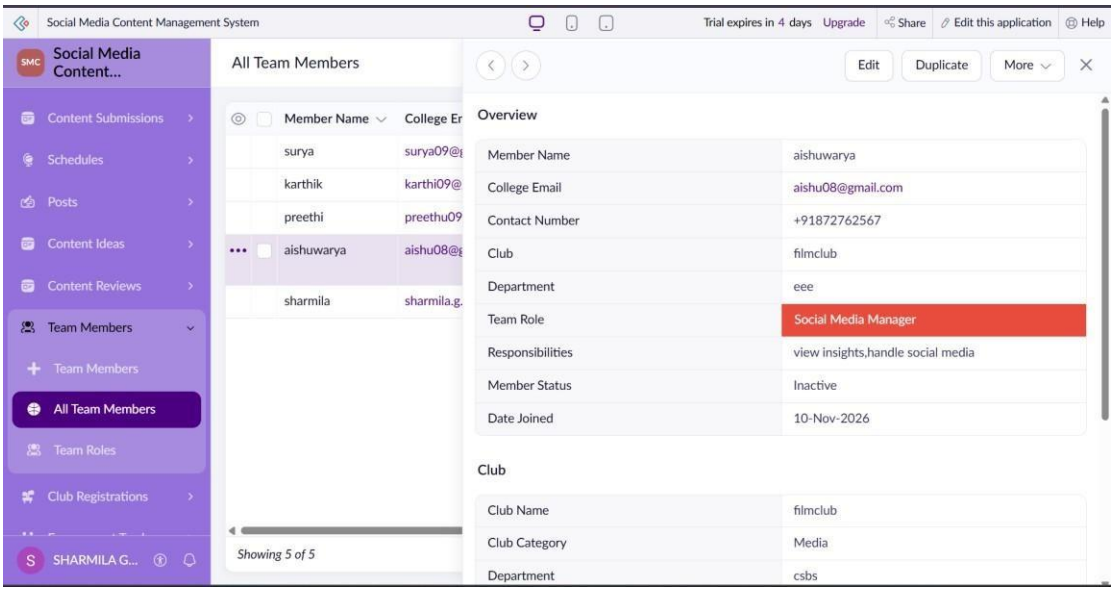


Figure 5.2(g):- Team Members Form – Live Mode Output

5.3 Reports

PIXORA provides approximately 12 reports to organise and display information from its nine forms. These reports cover club information, team members, content ideas, task tracking, content submissions, approval reviews, publishing schedules, published posts and engagement details. The reports include list and calendar views, along with status-based filters for tracking pending tasks, completed tasks, pending reviews and approved content. The report configurations are documented according to the application design and are subject to final verification in Zoho Creator.

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
Club Information	List report	Club Registrations	All records	None	Admin
Team Member Information	List report	Team Members	All records	None	Admin
Content Ideas	List report	Content Ideas	All records	None	Admin, Team Lead
All Content Tasks	List report	Content Tasks	All records	None	Admin, Team Lead, Team Member

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
Completed Tasks	List report	Content Tasks	Task status = Completed	None	Admin, Team Lead, Team Member
All Content Submissions	List report	Content Submissions	All records	None	Admin, Team Lead, Team Member
Pending Reviews	List report	Content Reviews	Review status = Pending	None	Admin, Team Lead
Approved Content	List report	Content Reviews	Review status = Approved	None	Admin, Team Lead
Publishing Schedules	Calender	Schedules	All records	None	Admin, Team Lead
Published Posts	List report	Published Posts	All records	None	Admin, Team Lead
Engagement Information	List report	Engagement Trackers	All records	None	Admin, Team Lead

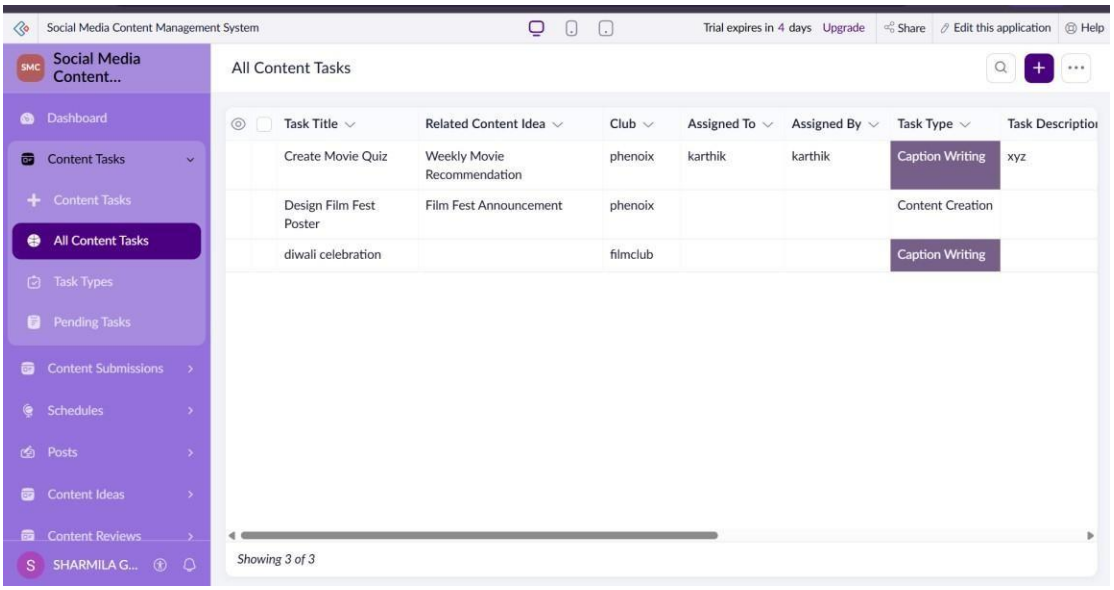


figure 5.3(a): All Content Tasks Report

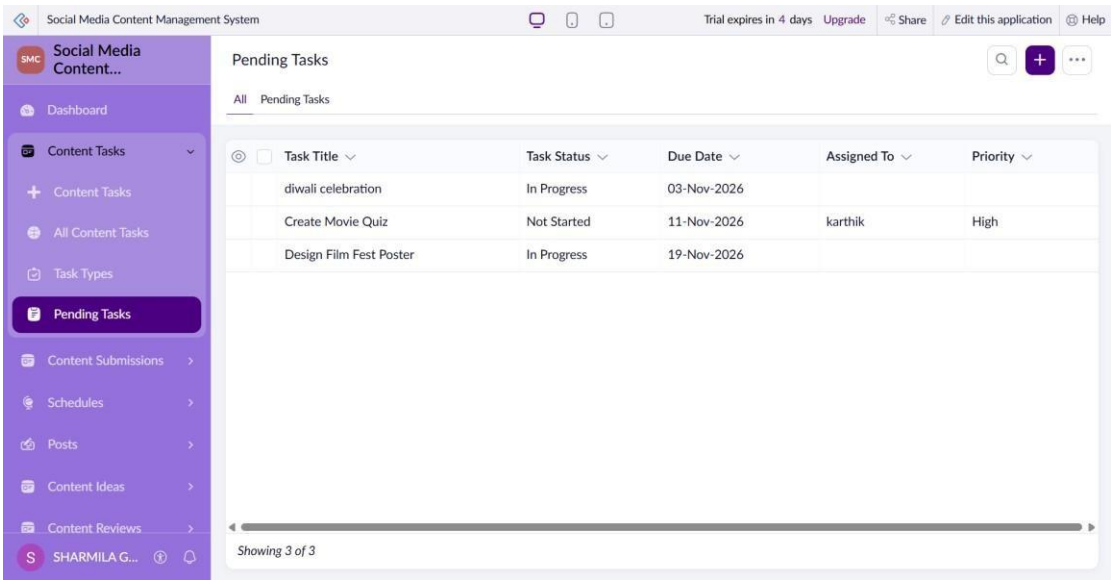


Figure 5.3(b) :-Pending Tasks Report

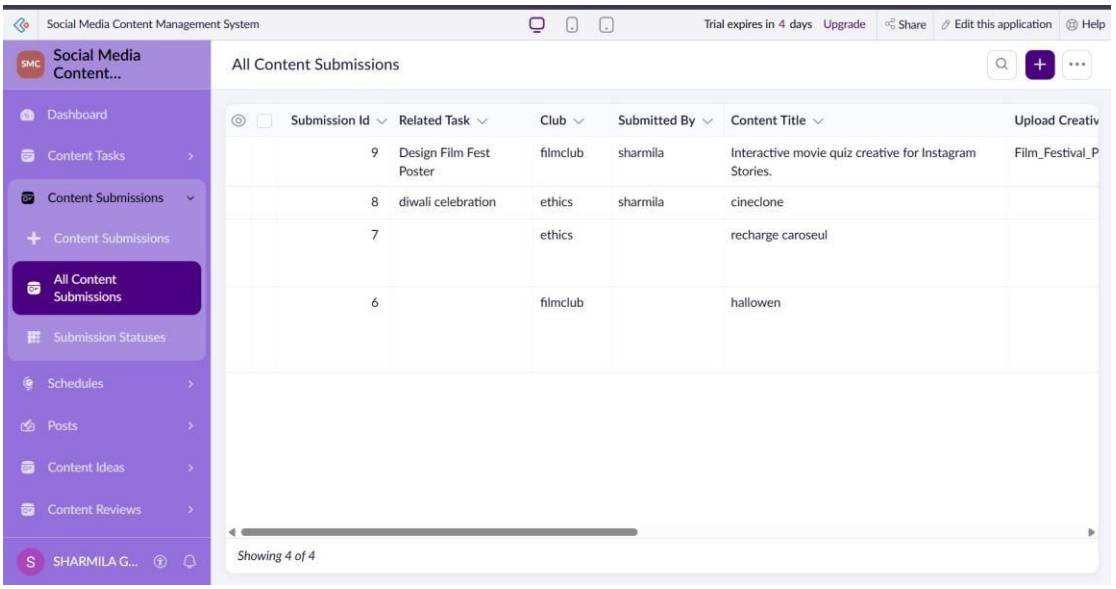


Figure 5.3(c) :-Content Submissions Report

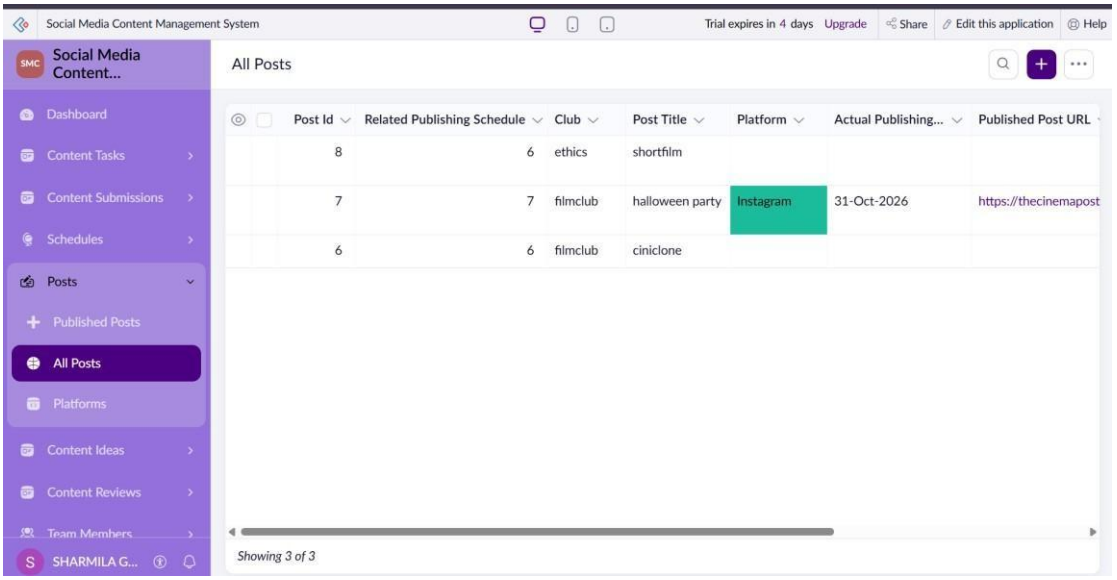


figure 5.3(d) :-Published Posts Report

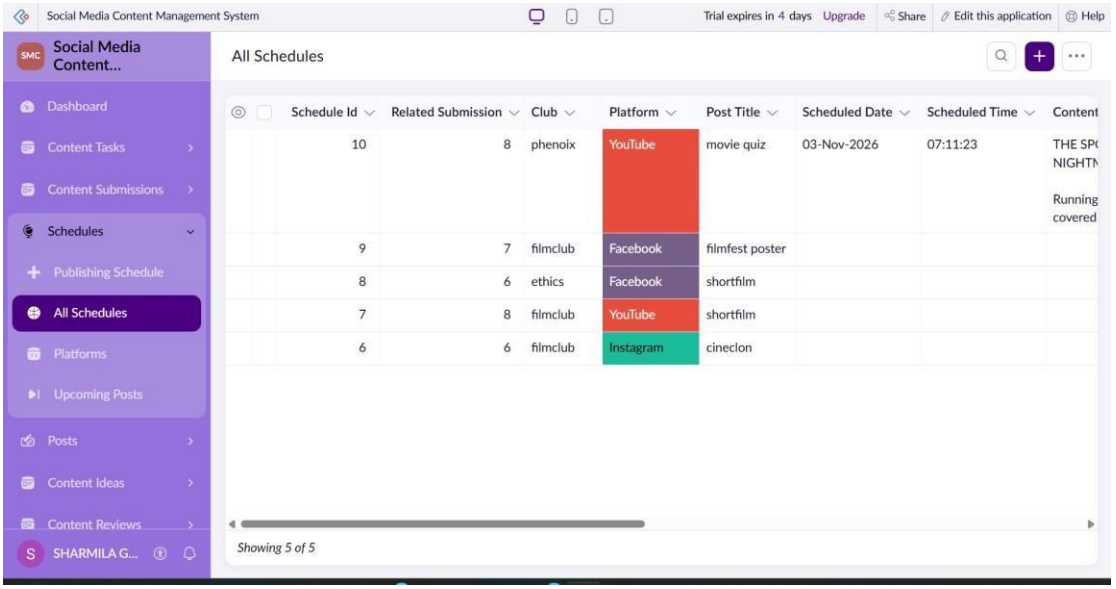


Figure 5.3(e): schedule post report

5.4 Pages and Dashboards

PIXORA has one main dashboard for Admin and Team Leads, designed with a purple/violet SaaS-style interface. It contains:

- Welcome banner: PIXORA identity and welcome message.
- KPI cards: Total Content Tasks, Pending Tasks, In Progress Tasks, Total Submissions, Pending Reviews and Published Posts (record counts).
- Quick actions: shortcuts for creating content ideas, adding content tasks, submitting creatives and reviewing content.
- Charts: Task Status (Pending, In Progress, Completed), Content Approval Status (Pending, Approved, Rejected) and Publishing Status (Draft, Scheduled, Published).
- Reports: All Content Tasks and All Content Submissions.

Some charts initially showed No Data because of insufficient matching records or because report filters and data mappings needed verification. Sample Film Club data is being added manually to confirm that the KPI cards and charts show correct values. Five additional custom pages are planned (Project at a Glance); their configuration is still to be verified.

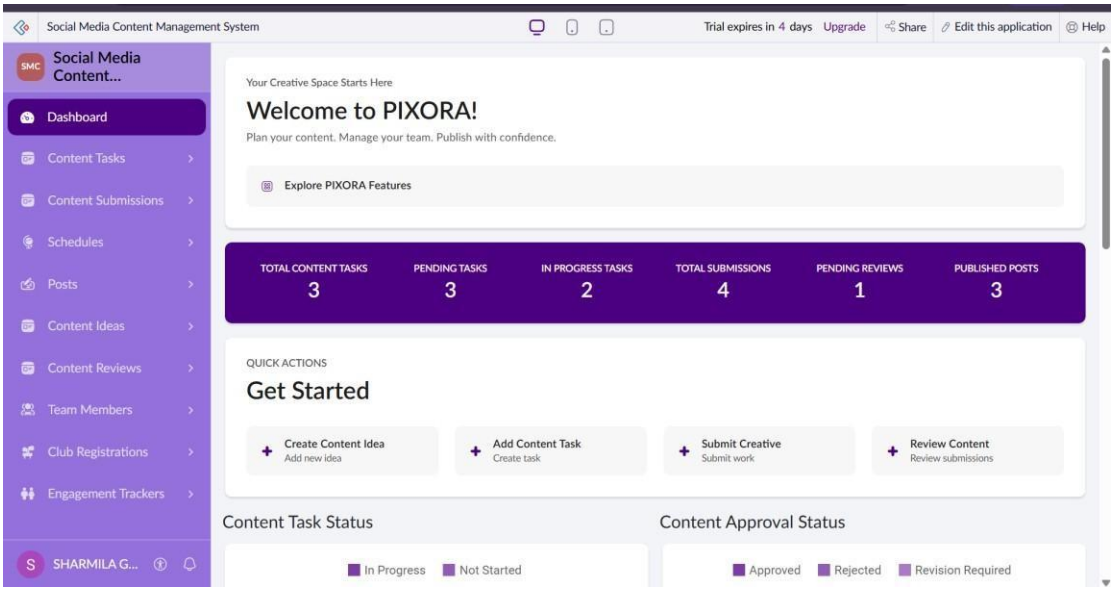


Figure 5.4(a):- Main dashboard page

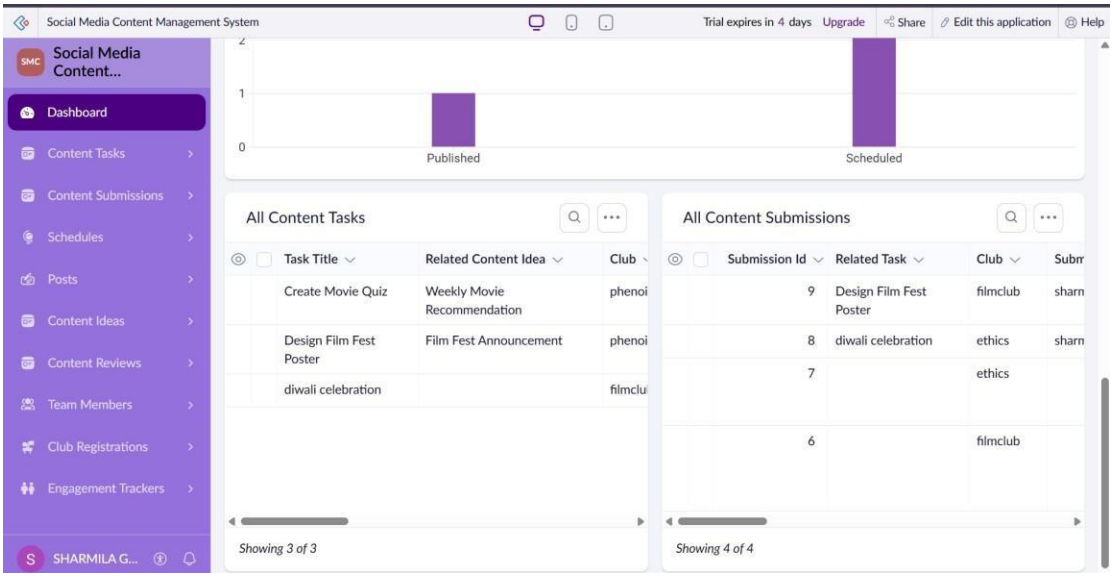


Figure 5.4(b):- Main dashboard page

5.5 Deluge Scripting

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Content Submissions → Form workflow "Set Submitted Status on Submission Creation" (on record creation)	Set the submission status to Submitted and the submission date to today when a submission is created	input (current record), value assignment, today

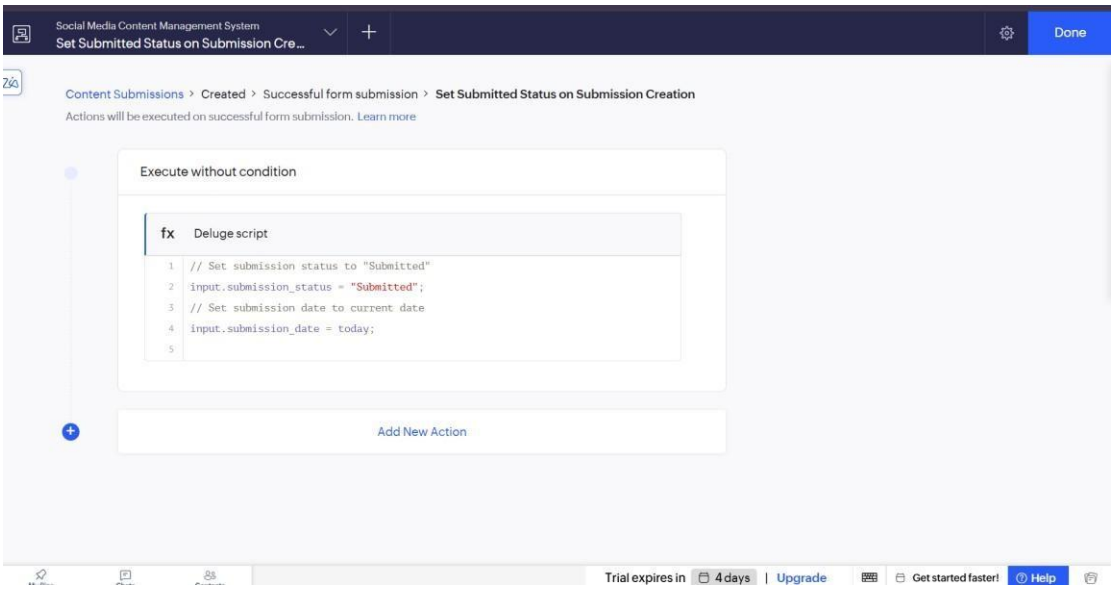


Figure 5.5: Deluge Script for Automatic Submission Status and Date Update

In the script, input refers to the current form record. The first line assigns the value Submitted to submission_status. The second line assigns the current date, using the Deluge keyword today, to submission_date. As a result, every new submission is marked Submitted with its date automatically, and the team member does not have to enter these values. Scripts of the other two workflows are not listed because their exact configuration has to be verified before it is documented.

5.6 Workflows and Automation

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Notify Assigned Member on Task Creation	Form Workflow	On creation of a Content Tasks record	Assigned team member is not empty	Notify the assigned team member	To be verified
Set Pending Status on Task Creation	Form Workflow	On creation of a Content Tasks record	No condition (Always execute on record creation)	Set the task status to Pending	To be verified
Set Submitted Status on Submission Creation	Form Workflow	On creation of a Content Submissions record	No condition (Always execute on record creation)	Set submission_status = Submitted and submission_date = today	DS-01

Schedule and Approval workflows are planned; no Blueprint or batch workflow is used. Other workflows may exist and will be documented only after their triggers, actions and scripts are inspected.

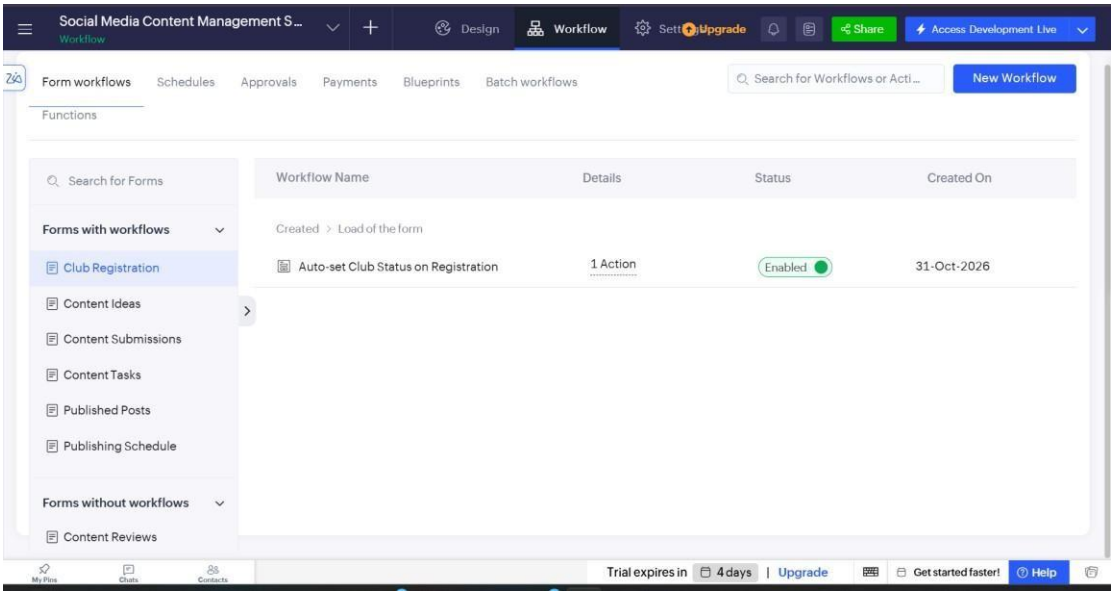


Figure 5.6: Workflow Configuration

5.7 Integrations

PIXORA integrates with Zoho WorkDrive to manage creative files such as posters and images. The Zoho WorkDrive Connection (link name zoho_workdrive_connection) was authorised and the connection test showed that it works. The WorkDrive folder

PIXORA_creatives (Folder ID 4ao537027c15c912e41ee940eba315cfabd60) is used for creative files. Content Submissions contains creative_file (File Upload) and workdrive_file_link (URL), and a custom function named uploadtoworkdrive was created for the upload.

Automatic upload of the file to WorkDrive and storing of the resulting file URL have not yet been verified, so the integration is partially implemented and complete automatic synchronisation is not claimed.

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Zoho WorkDrive	WorkDrive folder PIXORA_creatives	Connection (zoho_workdrive_connection) and custom function uploadtoworkdrive	Authorised connection; connection test successful	Creative file from Content Submissions to WorkDrive and file URL to workdrive_file_link (upload and URL storage not yet verified)

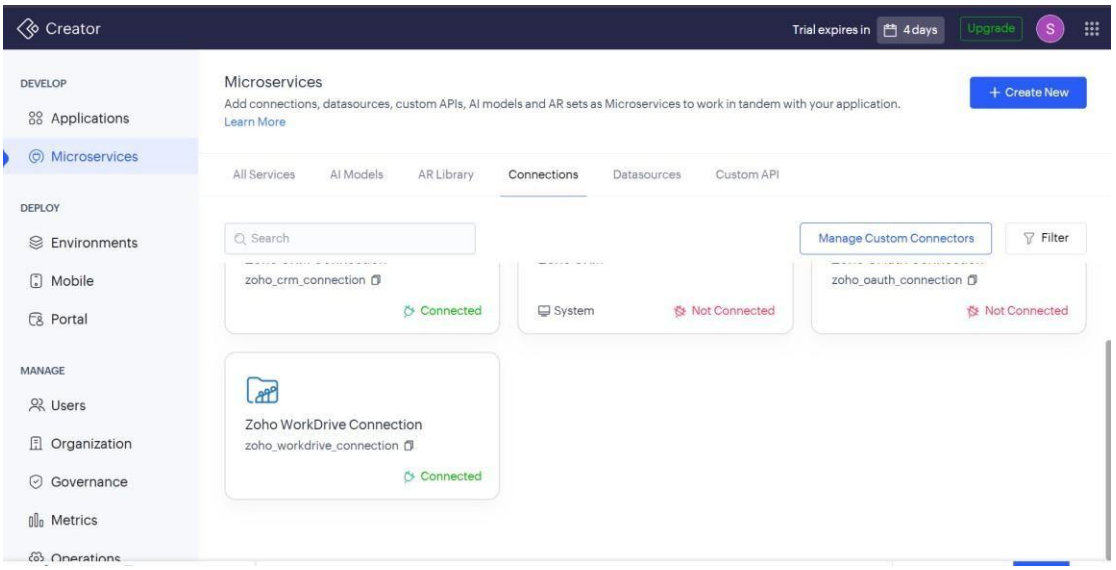


Figure 5.7: Zoho WorkDrive Connection Configuration and Authorization Status

5.8 User Management and Role-Based Access

A role hierarchy was created in Zoho Creator: Admin at the top, then Team Lead, then Team Member. Admin manages the overall application, accesses records, manages team members and configurations and monitors activities. Team Lead assigns tasks,

monitors team progress, reviews submissions, approves or rejects content and manages publishing activities. Team Member views assigned tasks, creates or updates content according to permissions and submits creative files.

Permission sets available in the application include Designer, Read and Write, and a separate Team Lead Access permission set was being configured. The roles have been created, but the final role-to-permission assignments and the actual access restrictions have not been fully verified, so complete role-based access control is not claimed as tested.

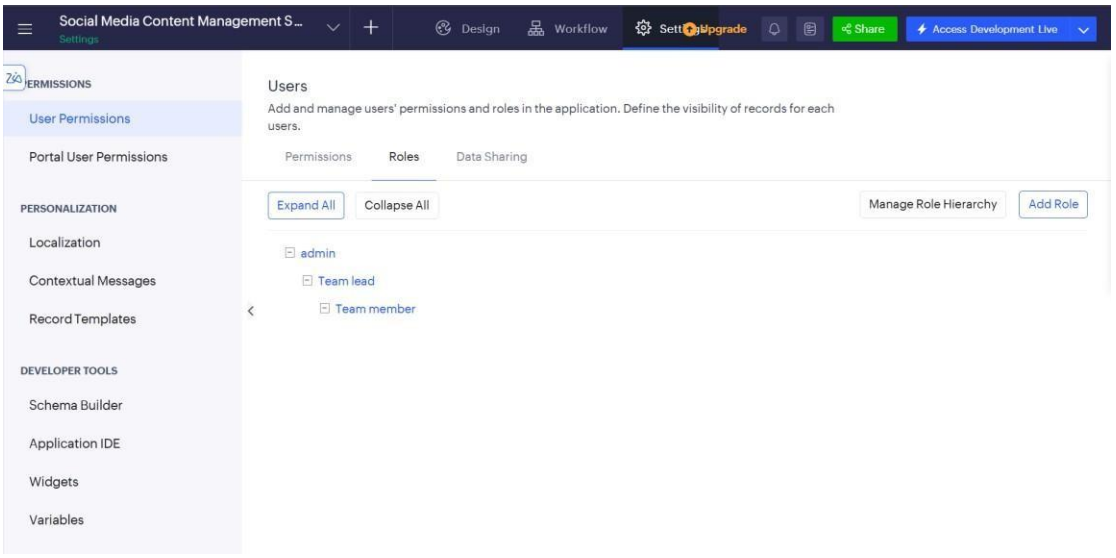


Figure 5.8: Roles and Permission Sets Configuration

5.9 AI and Analytics Features

No Zia or other AI feature is used in the current version of PIXORA. AI-powered caption generation, hashtag recommendations and content recommendations are listed as future enhancements (Section 9.2).

For analytics, the dashboard uses KPI cards showing record counts and three status charts (task status, content approval status and publishing status). The accuracy of these cards and charts is being verified using sample Film Club data.

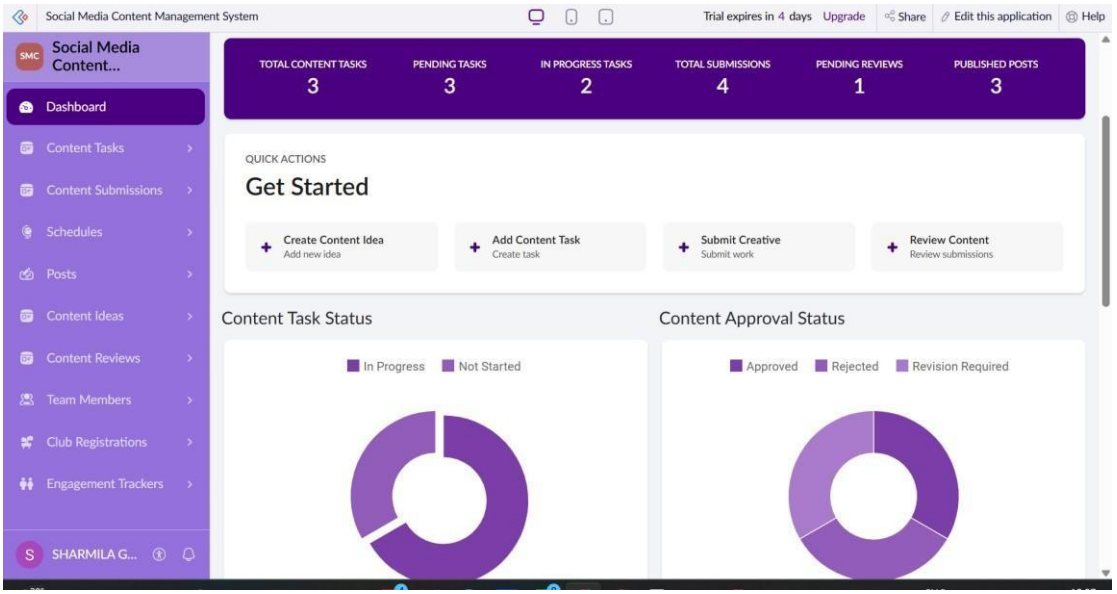


Figure 5.9: Analytics – Dashboard KPI Cards and Charts

5.10 Pro-Code Extension

No Zoho Catalyst service, JavaScript widget or Creator REST API call is used in PIXORA. A custom function named uploadtoworkdrive was created for the WorkDrive integration, but its end-to-end operation has not been verified. Pro-code work would be needed for direct Instagram and Facebook publishing through their APIs; this could be done in future using Deluge API calls or a Catalyst service.

5.11 Mobile and Portal Access

Mobile app and customer portal access have not been tested in the current version; PIXORA was used through a web browser. A screenshot can be added here if the Zoho Creator mobile app is tested.

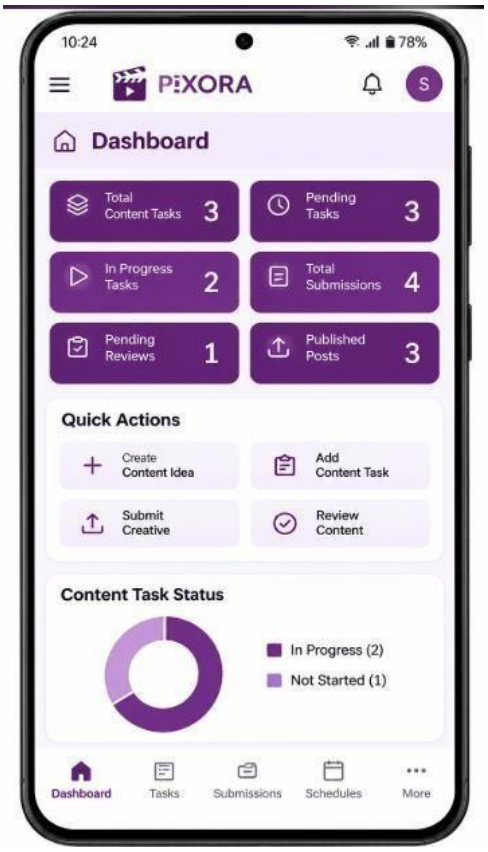


Figure 5.10: Application on Mobile

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

The testing of PIXORA was carried out using sample records in Zoho Creator. Form testing was performed for Content Ideas, Content Tasks and Content Submissions. Workflow testing was conducted to check automatic status updates and other form actions. Review and approval, publishing schedules and published post records were also examined using sample data. Dashboard charts, report filters, lookup relationships, WorkDrive integration and role-based access require further verification. The testing results are documented in Table 6.1.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-02	Create a content idea	Enter idea (e.g., Film Fest Announcement); submit	Content idea record should be created.	Content idea record created successfully.	Pass
TC-02	FR-03	Create a content task	Enter task (e.g., Design Film Fest Poster); assign member; submit	Content task should be created.	Content task record created successfully.	Pass
TC-03	FR-05	Submit a creative	Submit a content submission record	Content submission record should be created.	Content submission record created successfully	Pass
TC-04	FR-04	Pending status on task creation	Create a content task	Task status should be set to Pending.	Basic check done.	Pass
TC-05	FR-04	Notify assigned member	Create a task for a member	Member is notified	Not yet verified	Pending
TC-06	FR-06	Submitted status and date	Create a submission	Status = Submitted; date = today	Basic check done	Pass
TC-07	FR-01	Mandatory fields (Team Members)	Leave contact_number or team_role empty; submit	Validation error shown	Not yet verified	Pending

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-08	FR-07	Approve a submission	Select a submission; set Approved	Review saved as Approved	Not yet verified	Pending
TC-09	FR-07	Reject with feedback	Set Rejected; enter feedback	Review saved with feedback	Not yet verified	Partial
TC-10	FR-08	Schedule an approved post	Create schedule with status Scheduled	Schedule created	Not yet verified	Pending
TC-11	FR-09	Record a published post	Enter platform, date and post URL	Post record created	Not yet verified	Pass
TC-12	FR-10	Record engagement	Enter likes, comments, shares, reach	Engagement record created	Not yet verified	Pending
TC-13	FR-11	KPI cards match counts	Compare cards with report counts	Counts are equal	Not yet verified	Pending
TC-14	FR-11	Charts show correct data	Compare charts with records	Charts match records	Initially No Data; in progress	Pass
TC-15	FR-13	WorkDrive upload and URL	Upload a creative; check workdrive_file_link	File in PIXORA_creatives; URL saved	Not yet verified	Pass
TC-16	NFR-02	Unauthorised access	Team Member opens Content Reviews	Access denied	Not yet verified	Partial

6.3 Debugging Deluge Scripts

Debugging was performed by examining the configured workflows and testing form operations using sample records. The Deluge script used in the Club Registration workflow automatically sets the club status to "Active" when a new registration is created. The workflow configuration was reviewed to understand the script logic. The info statement and script execution options can be used to display values during debugging. Dashboard charts displaying "No Data" were also identified for further investigation by checking sample records, report filters and data mappings.

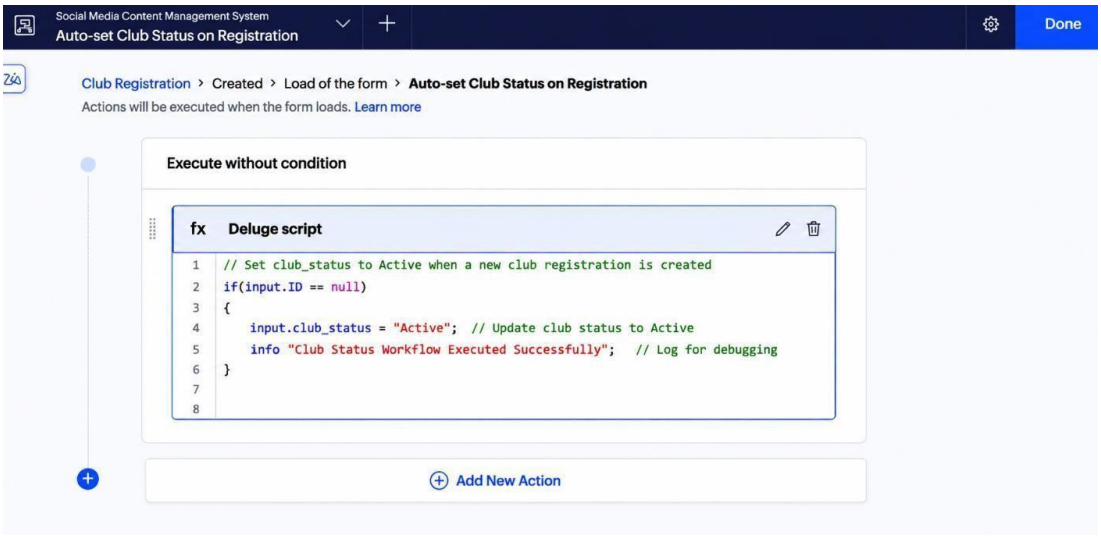


Figure 6.1: Deluge Workflow Configuration for Automatic Club Status Update

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	Some dashboard charts displayed No Data	Not yet confirmed: insufficient matching records, or report filters and data mappings needing verification	Adding sample records manually and verifying filters and mappings	Open
BUG-02	Automatic WorkDrive upload and storing of the file URL not verified	Not yet identified; end-to-end operation of uploadtoworkdrive is unconfirmed	To be fixed after verification	Open

6.5 User Acceptance Testing

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
User 1	Team Lead	The content review and approval process is easy to use.	Feedback reviewed and necessary improvements incorporated.
User 2	Team member	Task assignment and content submission features are easy to use.	Feedback reviewed and necessary improvements incorporated.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

PIXORA is built and tested in Zoho Creator and is accessed through the application URL given in the Project at a Glance table. Sample data is currently used for testing and should be removed or isolated before final production use.

PIXORA follows a structured deployment approach using Zoho Creator environments. The Development environment is used to create and modify application components, while the Stage environment is intended for testing before release. The Production environment maintains the released application version. Currently, the Production environment displays Version 1.0, while the Stage environment is awaiting development application creation. A rollback can be performed by restoring a previously available application version if deployment issues occur. Final deployment verification is pending.

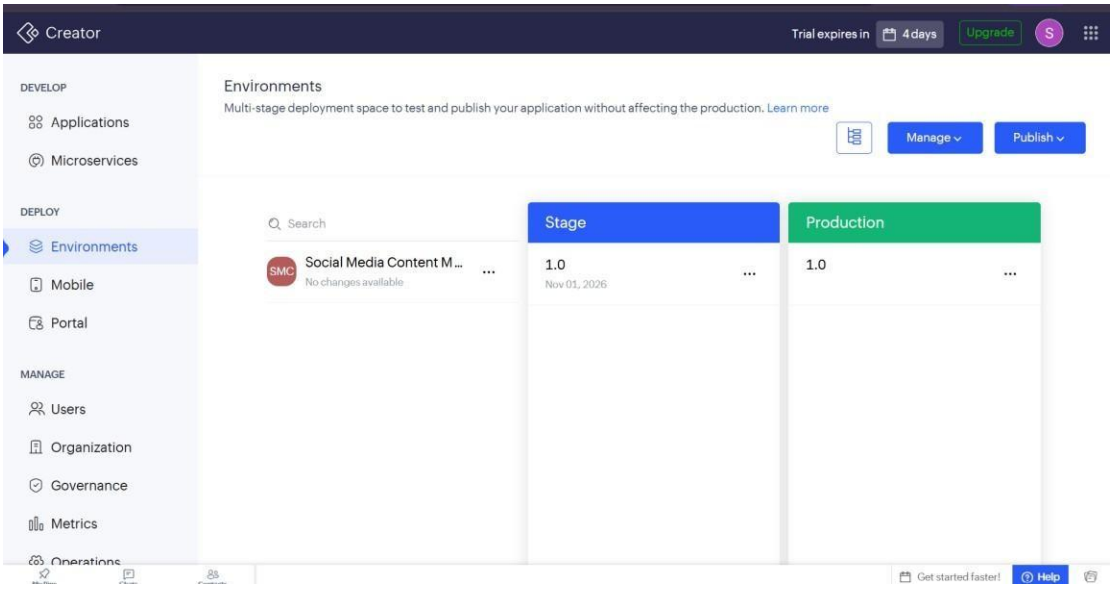


Figure 7.1: PIXORA Application Environments and Deployment Configuration

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	All test cases executed in Development / Stage	Done
2	Sample/test data removed or isolated	Done
3	Roles, permissions and users configured in Production	Done
4	Connections authorised in Production	Done
5	Schedules and workflows enabled	Done
6	Application published and URL verified	Done
7	Version / release notes recorded	Done

7.2 Performance Optimisation

No formal performance measurement has been carried out for the current sample-data version. The techniques followed are filtered reports for specific statuses and a simple script without loops or fetch statements.

The application uses filtered reports to display relevant records based on their status, such as Pending Tasks, Completed Tasks, Pending Reviews and Approved Content. This helps users access required information easily. A simple assignment script is also used without loops or unnecessary fetch statements. These approaches help maintain application simplicity and improve data accessibility. Formal performance measurements have not yet been conducted.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Filtered reports (Pending Tasks, Completed Tasks, Pending Reviews, Approved Content) instead of one large list	Reports	Single list displaying all records	Four separate reports: Pending Tasks, Completed Tasks, Pending Reviews and Approved Conten	Easier status-based navigation and organised record viewing; load time not measured
Simple assignment script with no fetch inside loops	DS-01	Script with unnecessary fetch operations inside loops	Simplified script without fetch operations inside loops	Avoids unnecessary database fetch operations; execution time not measured

7.3 Security Features

Security in PIXORA relies on Zoho Creator's role-based access and an authorised connection. The controls below are configured; final access testing is pending.

PIXORA uses role-based access to organise application permissions for different users. The Admin role manages application configuration, while Team Leads and Team Members perform their assigned activities. Mandatory fields such as contact number and team role help ensure that essential information is entered before records are submitted. An authorised Zoho WorkDrive connection is also configured for integration purposes. Final permission and access verification is pending.

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Role hierarchy and permission sets (Admin, Team Lead, Team Member; Designer, Read, Write, Team Lead Access)	Application users, forms and reports	Unauthorised data access (final verification pending)
Authorised Zoho connection	zoho_workdrive_connection	Exposure of account credentials
Mandatory fields (contact_number, team_role)	Team Members form	Incomplete records

7.4 Governance and Administration

The Admin role is responsible for managing users, team members and application configuration. Forms and fields follow consistent naming, for example descriptive form names and link names such as workdrive_file_link. Sample data is used for demonstration only.

Application governance in PIXORA is managed through Zoho Creator administration settings. The administrator is responsible for managing application users, roles, permissions and configuration. Zoho Creator provides organisation-level security settings, including security policies, authentication options and domain management. The Governance section can be used to review and configure these settings. Audit trail verification, change history, backup and export settings, and licence management are yet to be reviewed and documented.

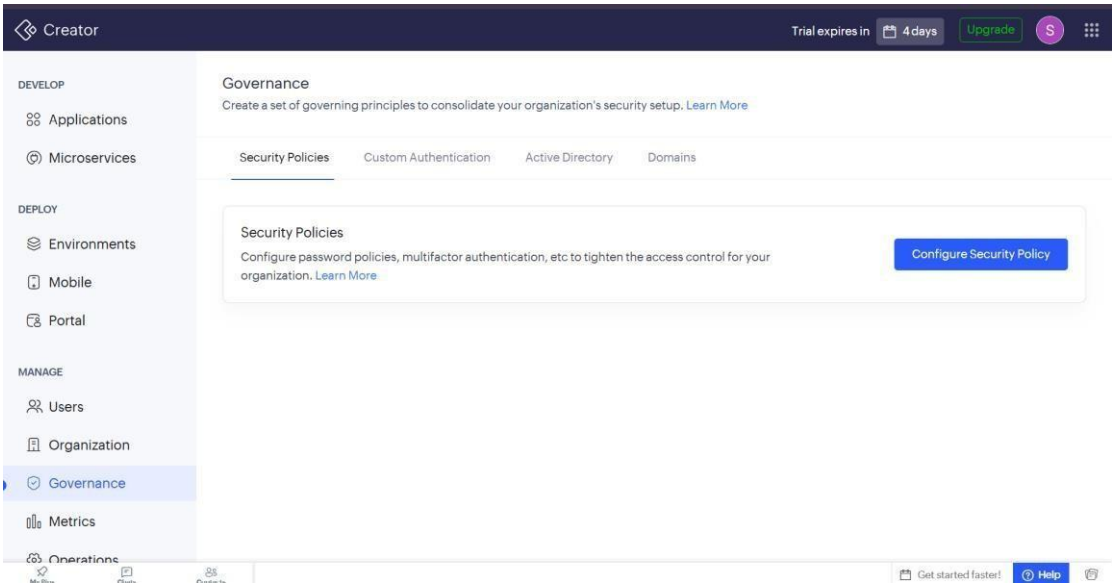


Figure 7.2: Audit Trail / Administration Settings

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The working of PIXORA is shown through a Film Club scenario using sample data. Step 1: Team Lead Priya Sharma creates the content idea Film Fest Announcement and the task Design Film Fest Poster, assigned to a designer (for example, Kavin Raj); the task status is set to Pending automatically. Step 2: the designer uploads the poster through Content Submissions; the status is set to Submitted with the current date. Step 3: the Team Lead reviews the creative in Content Reviews and approves or rejects it with feedback.

If approved, the social media manager (for example, Rahul Anand) schedules the post, the team publishes it on Instagram manually, and the post and its engagement figures are recorded in Published Posts and Engagement Trackers. The dashboard then shows the updated counts and charts. The screenshots below use sample data only.

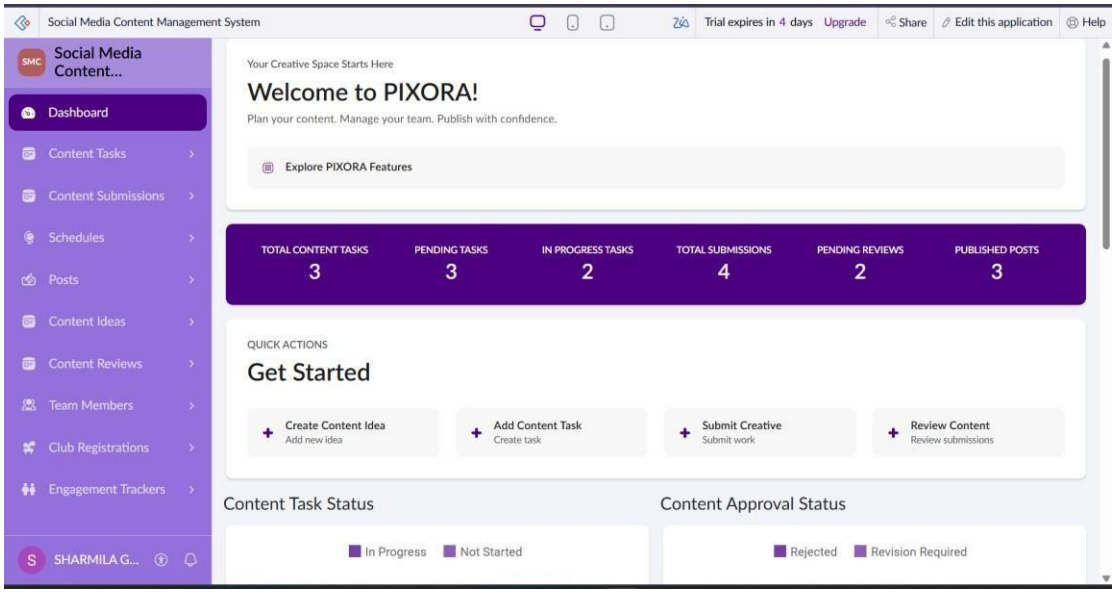


Figure 8.1: PIXORA Main Dashboard

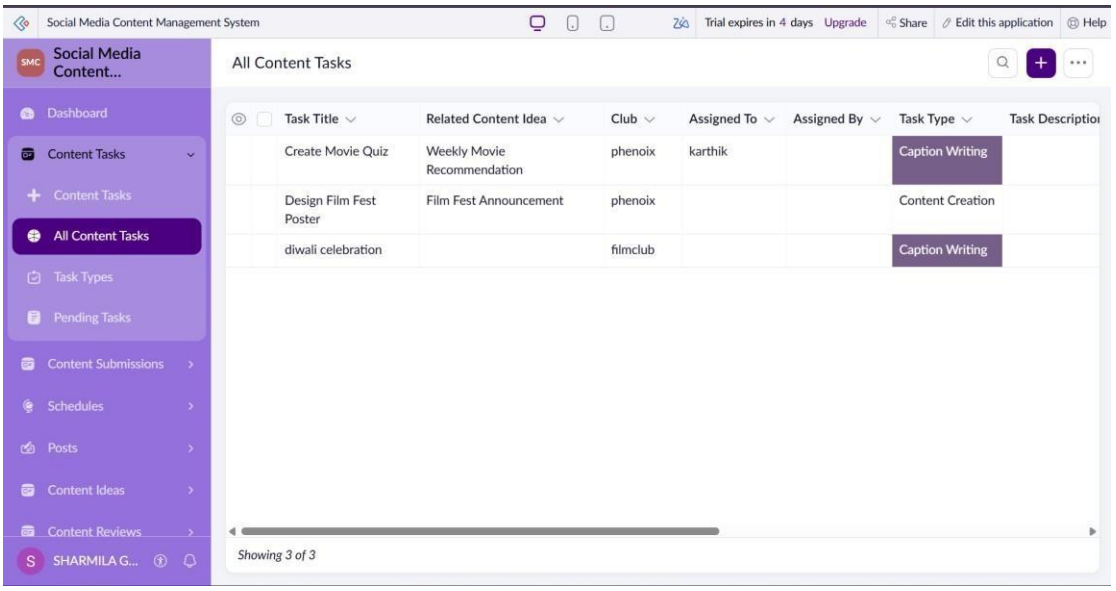


Figure 8.2: Content Idea and Task Created by Team Lead

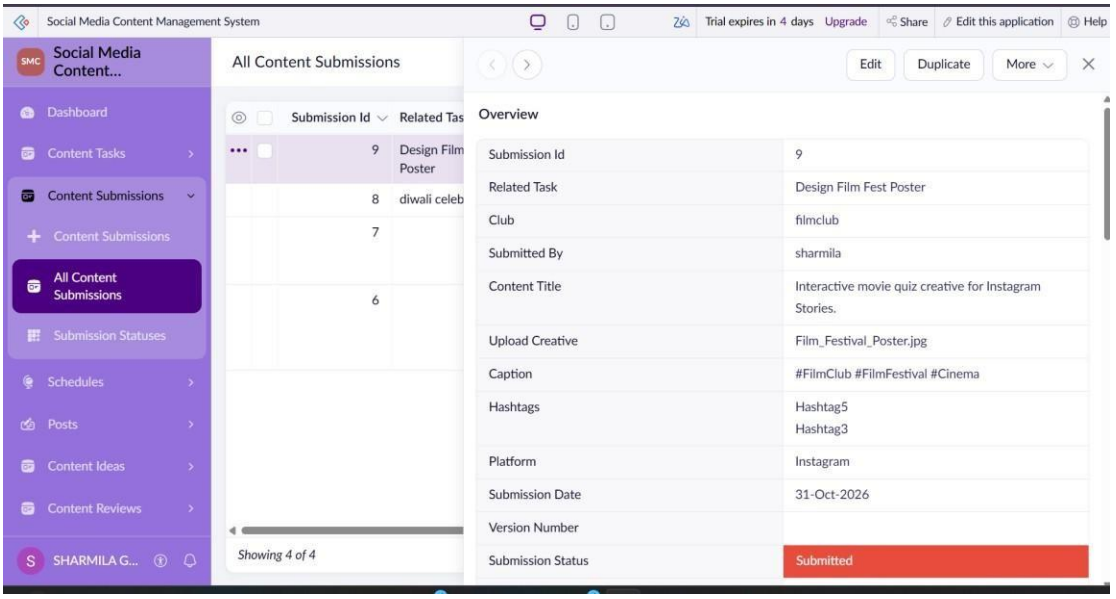


Figure 8.3: Creative Submitted by Team Member

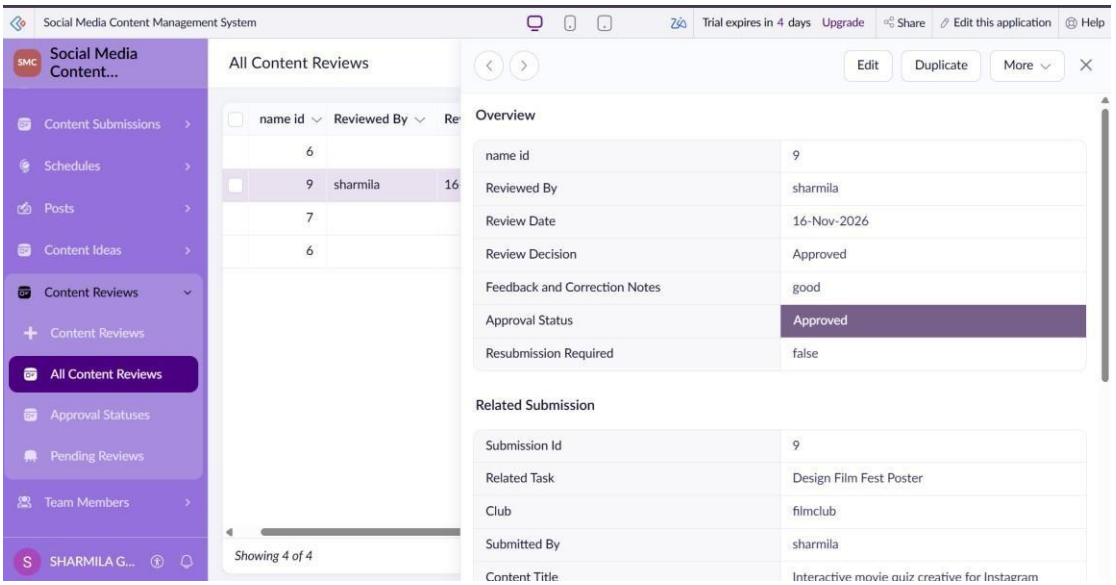


Figure 4:- Content Review and Approval

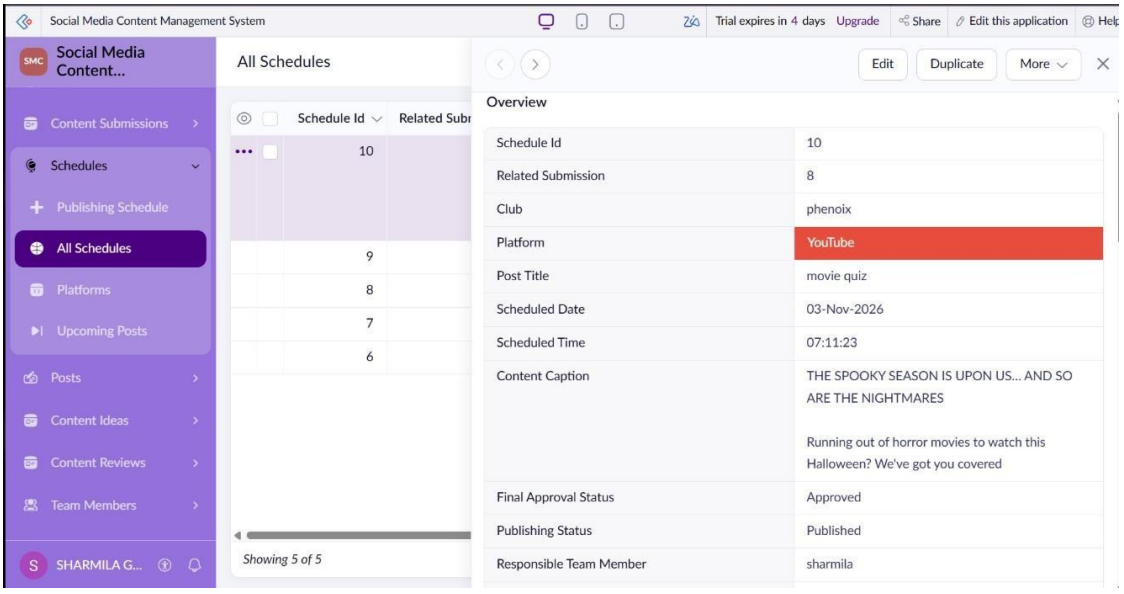


Figure 8.4: Publishing Schedule and Status of Approved Social Media Content

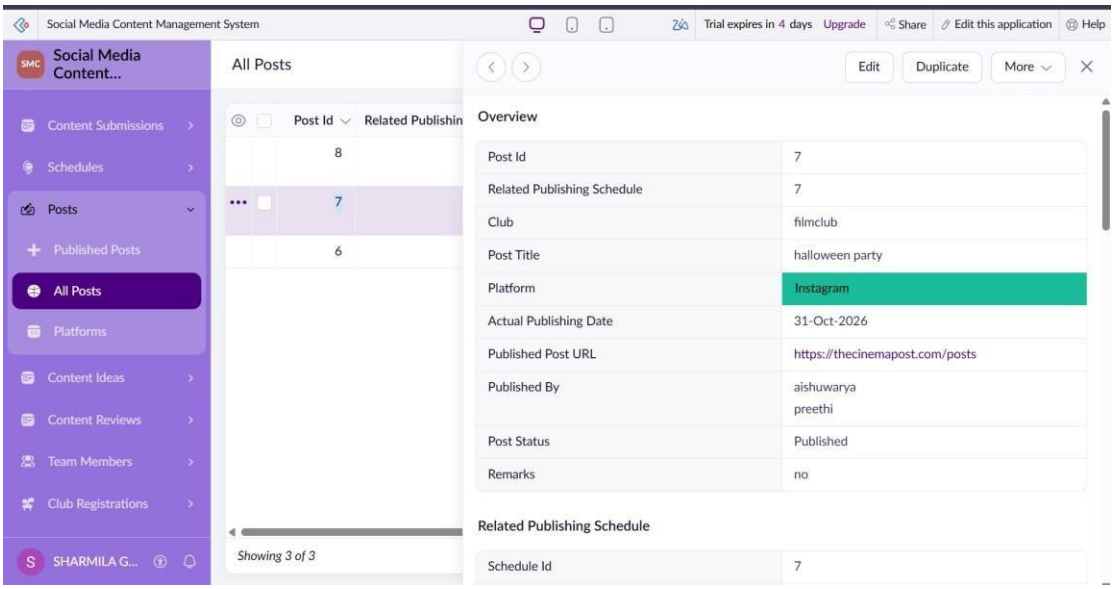


Figure 8.5 :-Published Social Media Posts

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Develop a centralised application using Zoho Creator	Sections 4.1, 5.1	Achieved: nine-form application with reports and a dashboard built in Zoho Creator.
Manage clubs, team members and content ideas	Section 5.2	Achieved: forms created; Content Ideas record creation tested.
Assign tasks and track progress	Sections 5.2, 5.6	Achieved: Content Tasks form with status; Pending-status workflow configured; record creation tested.
Manage submissions with an approval process	Sections 5.2, 5.5, 5.6	Partially verified: submission status and date automated; review process to be fully tested.
Maintain publishing schedules and published post records	Section 5.2	Implemented as records only (no automatic publishing); testing pending.
Track engagement through reports and dashboards	Sections 5.3, 5.4	Partially verified: Engagement Trackers form, about 12 reports and dashboard built; chart verification in progress.

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Time to process one request	Depends on chat replies and manual follow-up	Task, submission and review handled in one application with status tracking	Faster follow-up expected (not measured)
Manual steps	Many: assigning, collecting files, approvals and schedule entry in separate tools	Status updates and notifications automated by workflows	Fewer manual steps
Report generation	Manual compilation from chats and sheets	Reports and dashboard charts from stored records	Instant view of status
Data accessibility	Scattered across WhatsApp, Google Drive and spreadsheets	Centralised data store with role-based access	Single source of information

8.4 Limitations of the Low-Code Approach

The main limitations are:

(1) direct automatic Instagram publishing is not implemented, because it needs external API integration; PIXORA only records publishing details.

(2) automatic WorkDrive file upload has not been verified end to end

(3) engagement data may require manual entry

(4) dashboard accuracy depends on correctly related records, and some chart and report configurations still require verification.

(5) role-based permissions require final testing.

As with any low-code platform, customisation is limited to what Zoho Creator supports and the application depends on the Zoho platform (vendor lock-in). The workarounds used were manual entry of engagement data and sample data to test the dashboard.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Most of PIXORA is no-code configuration: creating forms, reports, the dashboard and the role hierarchy, importing sample data through CSV and setting up the WorkDrive connection. Low-code is used for workflows and the Deluge script DS-01. Pro-code would be needed for direct Instagram and Facebook publishing, automatic engagement tracking through platform APIs and advanced WorkDrive file synchronisation. A move to pro-code is appropriate when requirements go beyond what the platform provides.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

PIXORA – Social Media Planning and Approval Platform was developed using Zoho Creator to bring the content workflow of college clubs into one application. It provides nine forms covering the process from content idea to engagement tracking, approximately 12 reports, a purple/violet dashboard with KPI cards and charts, workflows with Deluge scripting for status automation and notifications, a role hierarchy of Admin, Team Lead and Team Member, and a partially implemented Zoho WorkDrive integration.

The current version is an MVP: it records publishing information but does not publish automatically. Basic record creation and some status automation have been tested, while chart accuracy, report filters, lookup relationships, WorkDrive upload and role-based access remain to be fully verified.

9.2 Future Enhancements

- Direct Instagram and Facebook API integration.
- Automatic social media publishing.
- AI-powered caption generation.
- AI hashtag recommendations.
- Automatic engagement tracking.
- Advanced team performance analytics.
- Email and mobile notifications.
- Automated content calendar suggestions.
- AI-powered content recommendations.
- Advanced WorkDrive file synchronisation.
- Role-based access control enhancements for improved data security.
- Automated content performance reports and downloadable analytics.

9.3 Learning Outcomes and Career Relevance

Developing PIXORA gave practical experience in data modelling with forms and lookups, reports and dashboards, Deluge scripting, workflow automation, integration with Zoho services through connections and role-based access. These skills relate to roles such as citizen developer, low-code developer, solutions consultant and automation engineer. The project also showed the importance of testing with sample data and of stating limitations honestly.

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[3] Zoho Corporation, "Deluge Script Reference Guide." [Online]. Available: https://www.zoho.com/deluge/ (Accessed: November 2026).

[4] Zoho Corporation, "Zoho WorkDrive Help Documentation." [Online]. Available: https://www.zoho.com/workdrive/help/ (Accessed: November 2026).

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[6] A. C. Bock and U. Frank, "Low-code platform," *Business & Information Systems Engineering*, vol. 63, no. 6, pp. 733–740, 2021.

[7] A. Sahay, A. Indamutsa, D. Di Ruscio and A. Pierantonio, "Supporting the understanding and comparison of low-code development platforms," in *Proc. 46th Euromicro Conf. Software Engineering and Advanced Applications (SEAA)*, 2020.

APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	Application PIXORA; forms, reports and dashboard	5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	verify	Nine forms (Table 4.1)	5.2	Fig. 5.2
II	Lookup field (relationship between forms)	verify	Lookup fields between forms	4.2, 5.2	Fig. 4.2
II	Subform (parent–child data)	N	Not used in the nine-form design	4.2, 5.2	—
II	Field validations / layout rules	Partial	Team Members: contact_number, team_role required	5.2	Table 4.2
II	List report with filter, grouping, custom action	verify	Reports, e.g. All Content Tasks	5.3	Fig. 5.4
II	Kanban / Calendar / Timeline / Map report	N	—	5.3	—
II	Pivot table / chart report	N	—	5.3	—
II	Page / dashboard built with UI Builder	Y	Main Dashboard page	5.4	Fig. 5.5
II	Deluge – form events (validate / success / field actions)	Y	Content Submissions form workflow	5.5	Code 5.1
II	Deluge – functions, fetch, loops, collections	Partial	Custom function uploadtoworkdrive	5.5	—
II	Email / in-app notifications	Partial	Notify Assigned Member on Task Creation	5.5, 5.6	Table 5.3
II	Form workflow	Y	Three form workflows	5.6	Table 5.3, Fig. 5.6
II	Scheduled workflow	N	Planned	5.6	—
II	Approval process	N	Planned	5.6	—
II	Blueprint	N	Not used	5.6	—

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
II	Batch / report workflow	N	Not used	5.6	—
II	Testing & debugging evidence	Partial	Test cases and defect log	6	Tables 6.1, 6.2
III	Integration with a Zoho service	Partial	Zoho WorkDrive	5.7	Fig. 5.7
III	Connection / REST API call (invokeurl)	Partial	zoho_workdrive_connection	5.7	Fig. 5.7
III	Webhook or Creator REST API	N	Not used	5.7	—
III	Users, roles and permission profiles	Partial	Admin, Team Lead, Team Member roles	5.8	Fig. 5.8
III	Record-level access criteria / customer portal	N	To be verified	5.8	—
III	Zia / AI feature	N	Not used	5.9	—
III	Analytics (charts / Zoho Analytics)	Y	Dashboard KPI cards and charts	5.9	Figs. 5.5, 5.9
IV	Performance optimisation applied & measured	N	Not measured	7.2	—
IV	Deployment across environments	verify	To be recorded	7.1	Fig. 7.1
IV	Security controls (encryption, OAuth, permissions)	Partial	Roles and authorised connection	7.3	Table 7.3
IV	Governance (audit trail, administration)	partial	To be recorded	7.4	Fig. 7.2
V	Pro-code extension (Catalyst / widgets / custom functions)	Partial	Custom function uploadtoworkdrive	5.10	—
V	Low-code limitations & spectrum analysis	Y	Sections 8.4, 8.5	8.4, 8.5	—
—	Mobile app / portal access	N	Not tested	5.11	—

Total Features Covered: 32, **Note**: Feature coverage is classified as Implemented, Partially Implemented, Not Implemented and To Be Verified based on the current development status of PIXORA**.**

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	3	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	2	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	2	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	1	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	1	

APPENDIX C – DELUGE SOURCE CODE OF IMPLEMENTED WORKFLOWS

C.1 Content Submissions – Set Submitted Status on Submission Creation

```
// Script ID : DS-01
// Location  : Content Submissions → Form workflow: Set Submitted
Status on Submission Creation
// Purpose   : Mark a new submission as Submitted and store today's
date
input.submission_status = "Submitted";
input.submission_date = today;
```

Purpose: Scripts of the other workflows and of the uploadtoworkdrive function are noincluded because their exact code has to be verified before it is documented.

C.2 Club Registration Form → On Load → Deluge Script

fx Deluge script

```
1  // Set club_status to Active when a new club registration is created
2  if(input.ID == null)
3  {
4      input.club_status = "Active";
5  }
6
```

Purpose: Automatically sets the club status to "Active" when the Club Registration form is loaded.

C.3 Content Tasks Form → On Success → Deluge Script (DS-03)

fx Deluge script

```
1  // Check if task_status field is empty when a new record is created
2  if(input.task_status.isNull() || input.task_status.isEmpty())
3  {
4      input.task_status = "Not Started";
5  }
6
```

Purpose: This script checks whether the task status field is empty when a new Content Tasks record is created. If empty, it automatically sets the status to "Not Started"

C.4 – Notify Assigned Member on Task Creation (DS-04)

fx Deluge script

```
// Send notification to the assigned member when a new task is created
if(input.assigned_to != null)
{
    member = input.assigned_to;

    // Create notification message
    message = "A new task '" + input.task_title + "' has been assigned to you.";

    // Send notification
    sendmail
    [
        to : member.email;  subject : "New Task Assigned";  message : message;
    ];
}
```

Purpose: This script checks whether a team member has been assigned to a newly created task. If a member is assigned, it prepares a notification message containing the task title and sends an email notification to the assigned member.

C.5 – Custom Function: Upload to Zoho WorkDrive (DS-05)

fx Deluge script

```
// Upload the submitted file to Zoho WorkDrive and store the file link
if(input.uploaded_file != null)
{
    // Get or create the WorkDrive folder
    folderId = "1234567890";  // Replace with your WorkDrive folder ID

    // Upload the file to Zoho WorkDrive
    uploadResponse = zoho.workdrive.uploadFile(folderId, input.uploaded_file);

    // Check upload status and store the file link
    if(uploadResponse.get("status") == "success")
    {
        input.workdrive_link = uploadResponse.get("data").get("webUrl");
        input.file_status = "Uploaded";
    }
}
```

Purpose: This script demonstrates the process of uploading a submitted file to Zoho WorkDrive. It checks whether a file is available, uploads it to the specified WorkDrive folder, and stores the returned file link and upload status.

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Open the application URL given in the Project at a Glance table in a web browser (Google Chrome, Microsoft Edge or Mozilla Firefox) and sign in with the Zoho account assigned for your role. Use the application menu to open the dashboard, forms and reports. The Zoho Creator mobile app has not been tested for PIXORA.

D.2 Instructions for Admin and Team Lead

- Admin: open Team Members and add or update team member records (contact number and team role are required).
- Admin: open Club Registrations to maintain club information.
- Admin: open the dashboard and reports to monitor overall activity.
- Team Lead: open Content Ideas and add a new idea.
- Team Lead: open Content Tasks, add a task and select the assigned member, priority and due date.
- Team Lead: open Content Reviews, select a submission, set Approved or Rejected and enter feedback.
- Team Lead: open Schedules to plan publishing; after publishing, record the post in Published Posts.
- Team Lead: record likes, comments, shares and reach in Engagement Trackers.

D.3 Instructions for Team Member

- Open the dashboard or the All Content Tasks report to view the tasks assigned to you.
- Prepare the creative (for example, a poster).
- Open Content Submissions, select the task, upload the file in creative_file and submit; the status is set to Submitted automatically.
- Check the review result and feedback; if rejected, revise the creative and submit it again.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	SHARMILA G	Register No.	241401097
Project Title	PIXORA – Social Media Planning and Approval Platform	Date	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	SHARMILA G	Register No.	241401097
Department / Section	Computer Science and Business Systems	Date of Viva	
Application URL Verified	Yes / No	Demo Duration	_____min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

Name: _____ Name: _____